Learning Whom to Trust from Feedback: How History Format Shapes In-Context Learning in Vision-Language Models

Anonymous submission

Abstract

Learning whom to trust when an advisor's accuracy changes is a classic test of context-dependent learning in people. We adapt a human advice-taking task to ask whether vision-language models (VLMs), without parameter updates, learn advisor accuracy in context from trial-by-trial feedback, and how this depends on the format of the history. The same history is presented as RAW (reports and outcome labels), RAW+CORRECTNESS (adding who was correct), CORRECTNESS-ONLY (who was correct alone), or SUMMARY (each advisor's count of correct trials). On 48 held-out episode pairs, four of five open-weight VLMs answering directly chose at or below chance from RAW and tended to repeat the previous outcome. CORRECTNESS-ONLY raised prediction accuracy by 8–16 percentage points to values close to those with SUMMARY, whereas RAW+CORRECTNESS helped little; masking parts of the history showed that the visible outcome label drives the repetition. The effect depends on the system and the layout: two proprietary reasoning models, tested on a subset, had observed RAW accuracy similar to that of a Bayesian volatility learner (63%), and a compact one-line layout narrowed the SUMMARY−RAW gap in two open models. In a changing bandit from Psych-101, a dataset of human choices, participants' estimated forgetting increased with the probability of change, while model estimates changed little when replaying participants' histories. The task and analysis code will be released upon acceptance.

Introduction

People learn whom to trust from feedback. When two advisors disagree and the outcome is revealed afterward, people keep track of how often each advisor was correct, and they update this estimate faster when the advisor's accuracy is volatile (Behrens et al. 2008); advisor reputations form quickly from a track record (Yaniv and Kleinberger 2000; Bonaccio and Dalal 2006). Because this advice-taking task comes from human neuroscience, a model can be tested with the same design and analyzed with the same computational models. We use a simplified variant without human data; our direct comparison with people uses a second task with published human choices.

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Agents built on large language models (LLMs) and vision-language models (VLMs) face the same problem when they combine tool outputs, retrieved documents or messages from other agents whose reliability can change. Prior work asked whether LLMs weight sources by reliability when it is fixed (Ashkinaze et al. 2026), is inverted at test time (Zhou et al. 2026) or breaks after scripted failures (Chen 2026), and proposed memories that track reliability outside the model (Feng, Yang, and Poria 2026). We ask how models without parameter updates use trial-by-trial feedback across repeated advisor-accuracy reversals, and how history format and its components affect that behavior. This is context-dependent learning without parameter updates: each query receives the accumulated history, and the model's choices do not change later feedback.

The format of past evidence is known to matter. LLMs choose differently when outcome probabilities are described than when they are experienced (Ge, Zhang, and Vorobeychik 2026); summarized histories can improve exploration in bandits (Krishnamurthy et al. 2024), and evidence integration can show strong recency (Xue, Wilson, and Xiong 2026). These tasks do not require deriving advisor correctness by comparing conflicting reports with observed outcomes.

Learning whom to trust involves three computations: (i) judging which advisor was correct on each trial, (ii) integrating these judgments across trials, and (iii) using the estimate. Five VLMs receive the same history in formats that state these results to different degrees (Fig. 1), plus a condition describing advisor accuracy. Formats also change length and salience; we test which history formats support good choices without assuming separable internal stages. Our contributions are:

- An advice-taking evaluation adapted from human neuroscience, with text and image reports; task generation, prompts and analysis code will be released upon acceptance.
- Across five open VLMs answering directly, CORRECTNESS-ONLY recovers most of the SUMMARY−RAW gap, whereas a visible outcome label elicits repetition of the previous outcome, as predicted before a held-out masking test.
- A boundary condition: two proprietary reasoning models have observed RAW accuracy similar to that of a Bayesian

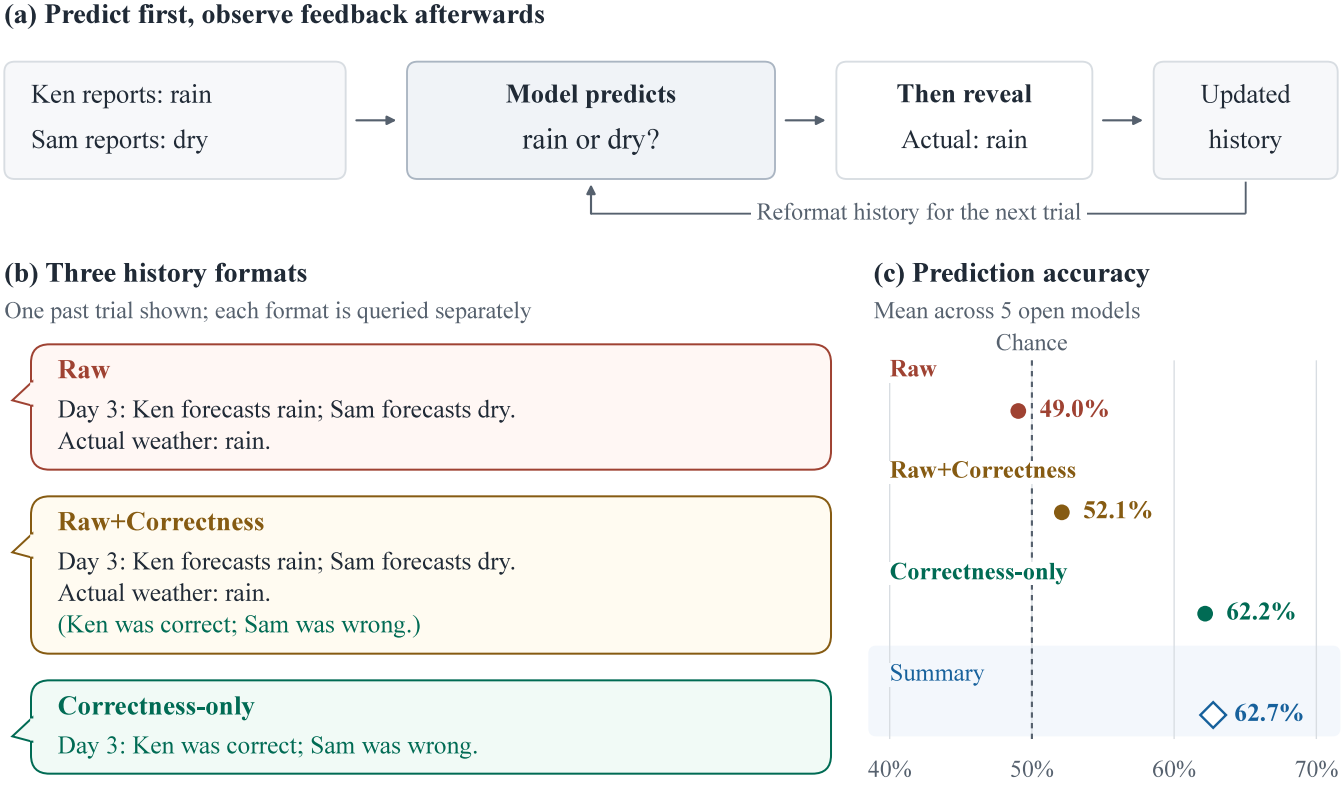


Figure 1: Replacing Raw with Correctness-only improves prediction accuracy. (a) Predictions use past history and current reports; the current outcome is revealed afterward and enters the next trial's history. (b) Raw lists reports and outcome labels; Raw+Correctness appends who was correct; Correctness-only retains only those correctness statements. Excerpts show the same past trial in separate queries. (c) Summary supplies each advisor's count of correct trials (open diamond). Markers show equal-weight mean accuracies over five open models on 48 held-out episode pairs with text reports. An episode pair comprises an episode and its counterbalanced copy, with advisor names swapped and the answer-word mapping reversed. The dashed line marks 50% chance. Table 1 gives model-level accuracies and paired 95% bootstrap confidence intervals (CIs) for Summary−Raw differences; Table 7 gives individual accuracy CIs.

volatility learner, so the history-format effect depends on the system.

- A descriptive human–model comparison in a Psych-101 changing bandit: participants' estimated forgetting rose with the hazard rate (the per-trial probability of a payoff change), whereas the open models' estimates changed little.

Related Work

Cognitive evaluation of LLMs. Treating a language model as a participant in a psychological experiment is now a common way to characterize its behavior (Binz and Schulz 2023). CogBench turns such experiments into behavioral metrics (Coda-Forno et al. 2024), and Centaur is a model fine-tuned to predict trial-by-trial human choices in Psych-101 (Binz et al. 2025). Fits of reinforcement-learning models to in-context behavior in bandits have found asymmetric or biased updating and rigid or greedy choice (Schubert et al. 2024; Hayes, Yax, and Palminteri 2024; Wang, Ward, and Zhang 2026b; Schmied et al. 2025), and reversal-learning schedules have been compared across LLMs (Wang, Ward, and Zhang 2026a; Li et al. 2025). The bandit and reversal-learning evaluations focus on learning option values rather than extracting advisor correctness from report–outcome pairs.

Description, experience and history summaries. The description–experience gap (Hertwig et al. 2004; Wulff, Mergenthaler-Canseco, and Hertwig 2018) has an LLM counterpart in the description–history gap of Ge, Zhang, and Vorobeychik (2026). Krishnamurthy et al. (2024) contrasted raw and summarized histories in bandits, and Xue, Wilson, and Xiong (2026) traced integration kernels (weights on evidence at different lags) in a counting task. LLMs also update roughly as Bayesian observers when the observed sequence is itself what they predict (Gupta et al. 2025; Zhang, Yang, and Wang 2025). Our Summary is computed from the same history the model would otherwise see; it discards trial order and is not a description. Source-history interventions also distinguish recognizing reliability from allowing it to influence a decision (Zhang, Jiang, and Lu 2026). We extend this

question to repeated reversals and test which components of trial-by-trial feedback trigger outcome repetition.

Volatility and multimodal reports. Our task follows the social-learning paradigm of Behrens et al. (2008). Normative accounts of learning under volatility include hierarchical Bayesian and change-point models (Behrens et al. 2007; Nassar et al. 2010, 2012; Mathys et al. 2011; Piray and Daw 2021, 2024; Soltani and Izquierdo 2019), and social and individual learning can be arbitrated separately (Diaconescu et al. 2020). For image reports, performance depends on modality even when content is isomorphic (Fu et al. 2024), multi-image inputs show position biases (Tian et al. 2025), and VLMs have difficulty binding features to objects (Campbell et al. 2024), while supplying explicit intermediate features can shape what a VLM outputs (Lee et al. 2026).

Task and Methods

Advice-taking task. On each trial, two advisors give conflicting binary reports (for example, rain and dry), the model predicts the outcome, which amounts to choosing whom to follow, and the outcome is then revealed. An episode has 80 trials. Advisor A, the advisor with the higher scheduled accuracy on trial 1, is correct with probability 0.8 in a stable phase of 40 trials; in a volatile phase of 40 trials its accuracy alternates between 0.8 and 0.2 every 10 trials (each switch is a reversal), following Behrens et al. (2008) (comparison in Appendix B). Half of the episodes start with each phase. Outcome sequences are pre-generated, so the model's choices do not change the history. Within each 10-trial block, A's realized accuracy equals its scheduled accuracy, and each outcome label appears five times. The instructions say that an advisor's accuracy may change but give no rate.

Each episode is shown with a pet cover story, in which advisors say whether a hidden animal is a cat or a dog, and with one of six non-pet cover stories (colors, weather, stocks, sports, medical tests, news; Appendix A). Current reports are given as text or as an image: a drawn icon for the non-pet cover stories and a photograph (Parkhi et al. 2012) for the pet cover story; with text reports, both cover stories are text only. Every episode has a counterbalanced copy in which advisor names are swapped and the answer-word mapping is reversed while the outcome sequence is unchanged; averaging over the two copies cancels advisor-name and answer-word biases. The 16 development episodes were used to choose settings; 48 separate episodes form the held-out set. A robustness set has 24 episodes with independent outcomes and irregular reversals.

History formats. Before trial t, the model sees trials $1, \ldots, t-1$ in one of four history formats (Fig. 1; full prompts in Appendix A); prompts name trials after the cover story (e.g. days). RAW lists both reports and the outcome label ("Day 3: Ken forecasts rain; Sam forecasts dry. Actual weather: rain."). RAW+CORRECTNESS appends "(Ken was correct; Sam was wrong.)" to each trial. CORRECTNESS-ONLY keeps only that correctness statement, stating the result of computation (i). SUMMARY gives each advisor's number of correct trials, stating the result of computations (i) and (ii); it discards trial order. SUMMARY variants count over the last K trials or since a change point inferred by Bayesian online change-point detection (BOCPD; Adams and MacKay 2007; Appendix E). A separate correctness chart presents trial-level correctness as an image. These format names refer to inputs given to the VLM. The *count rule* instead chooses the advisor with more correct trials over the corresponding range, without a VLM. Unless specified otherwise, SUMMARY and the count rule use all past trials.

Control conditions. On development episodes, control items give the answer more directly: *described accuracy* ("Ken was correct on 80% of past days"), described counts (counts of correct trials stated without the trial history), short and long easy histories in RAW and CORRECTNESS-ONLY formats, and a single RAW trial on which the model names who was correct (computation (i) alone).

Models and readout. We test five open-weight VLMs without fine-tuning, four of them built on Qwen3 language models: Qwen3-VL-8B and Qwen3-VL-32B (Bai et al. 2025), InternVL3.5-8B and InternVL3.5-38B, which use a Qwen3 language model (Wang et al. 2025), and Gemma-3-27B (Gemma Team 2025); B denotes billions of parameters. The open models are prompted to answer directly with one word. Two proprietary reasoning models are tested through application programming interfaces (APIs) on a subset; their choice is read from the generated answer (Appendix J). For the open models, the readout is the choice probability y_t for advisor A's report, from the logits (unnormalized output scores) of the two answer words at the first generated token, normalized over these two words; it is not a belief. For four models nearly all probability mass fell on the two answer words; Gemma-3-27B often began a sentence instead when given SUMMARY (Appendix J).

Measures and analysis. For the open models, prediction accuracy (hereafter accuracy) is the percentage of trials ($t \geq 2$) on which the higher-logit answer is correct, after averaging label-aligned logits over the two copies of each episode pair (the episode and its counterbalanced copy); exact ties receive half credit and chance is 50%. The *currently better advisor* has the higher scheduled accuracy on that trial; the *following probability* is the choice probability for that advisor. To describe behavior we fit a logistic regression of y_t on which advisor was correct k trials back, whose weights over k form an integration kernel (cf. Xue, Wilson, and Xiong 2026), plus a regressor for repeating the previous outcome, and compute chance-corrected agreement (Cohen's κ) with simple strategies; *choice agreement* is the percentage of non-tied trials on which the model and a count rule choose the same advisor. As a computational reference we run a hierarchical Bayesian volatility learner inspired by Behrens et al. (2007, 2008) (implementation in Appendix D). It receives who was correct on each trial and assumes conditionally independent Bernoulli observations, without modeling the fixed per-block counts described above. The unit of analysis is the episode pair (an episode and its counterbalanced copy), and, unless specified otherwise, 95% confidence intervals (CIs) use 10,000 bootstrap resamples of episode pairs. Differences

between accuracies are in percentage points. These CIs are pointwise: they are not adjusted for simultaneous coverage across estimates. The primary test, fixed before held-out evaluation, is the SUMMARY−RAW accuracy difference per model (Wilcoxon signed-rank test, Holm-corrected across models; required in both language-model families). Secondary tests and all other details are in Appendix C.

Human comparison. We replay the changing-bandit task of Xiong et al. (2023) from Psych-101: 30 participants, three change probabilities (hazard rates, h) and 100 trials per game. The hazard rate is the probability that a machine's payoff changes on each trial. The five open models see each participant's preceding choices and payoffs, with h hidden or stated. We fit an adapted myopic model-free learner to human choices and sampled model choices. Its forgetting rate ϕ moves the unchosen machine's estimated value toward the midpoint of the payoff range; we compare slopes of ϕ on h, paired by participant (Appendix H).

Results

Open models use described accuracy more reliably than RAW. With text reports, described accuracy or counts led the models to choose the advisor described as better on 92–100% of items (a different measure from main-task accuracy), and every model identified which advisor was correct on a single shown trial (Table 6; development episodes). The same information was used much less when it had to be learned. In an easy history presented as RAW, in which one advisor had been correct on every recent trial, models followed that advisor on 60–79% of items after 10 trials and on 53–70% after 80. CORRECTNESS-ONLY raised this to 100% for 10 trials, but only to 67–90% for 80 trials. The difficulty therefore emerges when using multiple feedback trials, and long histories remain hard even when correctness is stated explicitly.

CORRECTNESS-ONLY recovers most of the SUMMARY−RAW gap; RAW+CORRECTNESS helps less. On the held-out set, accuracy was 46.8–48.6% with RAW in four models and 54.9% [53.8, 56.0] in InternVL3.5-38B, the only model above chance, and 61.7–63.5% with SUMMARY (Table 1; individual accuracy CIs in Table 7). The SUMMARY−RAW gap was $+8.4$ to $+16.4$ percentage points and significant in every model (Holm-corrected $p < .001$; Table 1), so the primary test held in both language-model families. Scoring each counterbalanced copy separately preserved positive gaps in all five models ($+8.8$ to $+13.9$ percentage points; Appendix J). CORRECTNESS-ONLY attained 60.2–63.6% accuracy, close to SUMMARY, although it still leaves integration to the model. Integration was nonetheless imperfect: CORRECTNESS-ONLY remained below the last-five count rule (69.2%) and lagged after reversals (below). RAW+CORRECTNESS improved accuracy by only 1.1–5.4 percentage points over RAW, even though the model then had all the information of CORRECTNESS-ONLY. The reports and outcome labels therefore do more than add work: they interfere with using the statement next to them, which the next analyses explain. The size of the SUMMARY−RAW gap depends on the prompt wording: with a compact one-line record per trial, RAW accuracy rose to 55–56% while CORRECTNESS-ONLY and SUMMARY fell, shrinking the gap to 1–3 percentage points in the two 8B models; RAW accuracy stayed below the count rule in every wording (development episodes; Appendix I).

Under RAW, open models tend to repeat the previous outcome. The integration-kernel weights on past advisor correctness were near zero, while the weight on repeating the previous outcome was positive in every model (Fig. 8; integration kernel in Fig. 3). Repeating the previous outcome agreed best with the models' choices ($\kappa = .22$–$.32$ in four models and .13 in InternVL3.5-38B, whose choices also agreed weakly with the last-five count rule; Fig. 9). The 15-lag integration kernel describes local sensitivity to recent trials and cannot by itself separate a recency-weighted learner from an all-trial count rule on this schedule; identifiability checks are described in Appendix F. Because each 10-trial block contains each outcome label five times, repeating the previous outcome scores below chance (44.5% on the held-out set), which explains accuracy below 50%; on the robustness set, where outcomes are independent, RAW accuracy was near chance (within 3 percentage points of 50%; 50.7–52.0%) in four models. Under CORRECTNESS-ONLY, the integration kernel instead peaked at the most recent trial and decayed over the next few, and the weight on repeating the previous outcome vanished.

Masking identifies the outcome label as a trigger. We replaced parts of RAW+CORRECTNESS by a dash while retaining its layout. Development results motivated predictions for a five-model held-out replication (Appendix I). On the held-out set, all predictions written before the run held in all five models: replacing the outcome label helped by 2.7–6.4 percentage points and reduced the weight on repeating the previous outcome from .42–.87 to .08–.23, replacing the reports alone changed accuracy by at most 0.4 percentage points, but once the outcome label was replaced, also replacing the reports added 2.1–5.5 percentage points, and with both replaced accuracy came within 4 percentage points of CORRECTNESS-ONLY (Table 2; Appendix I). The visible outcome label drives the repetition; the reports matter once it is removed.

A computational reference for evidence integration. The Bayesian volatility learner, run on the same held-out set, made correct choices on 66.8% [66.1, 67.4] of trials, above the all-trial count rule (63.2%) and above every open model given RAW, CORRECTNESS-ONLY or SUMMARY (Table 1; Appendix B). It receives who was correct on each trial, so it benchmarks integration separately from reading the history, and it is not human performance. The last-five count rule, selected on development episodes with the same schedule, scored higher (69.2%), and only models given SUMMARY variants over the last five trials or since a change point, computed outside the model, reached the learner's accuracy (66.6–69.1%; Appendix E). With 80 trials, even this learner's implied learning rate rises only slightly in the volatile phase, too little to test volatility adaptation here (Appendix F).

Following the count rule from SUMMARY also reproduces its lag. Given SUMMARY, models chose the advisor with

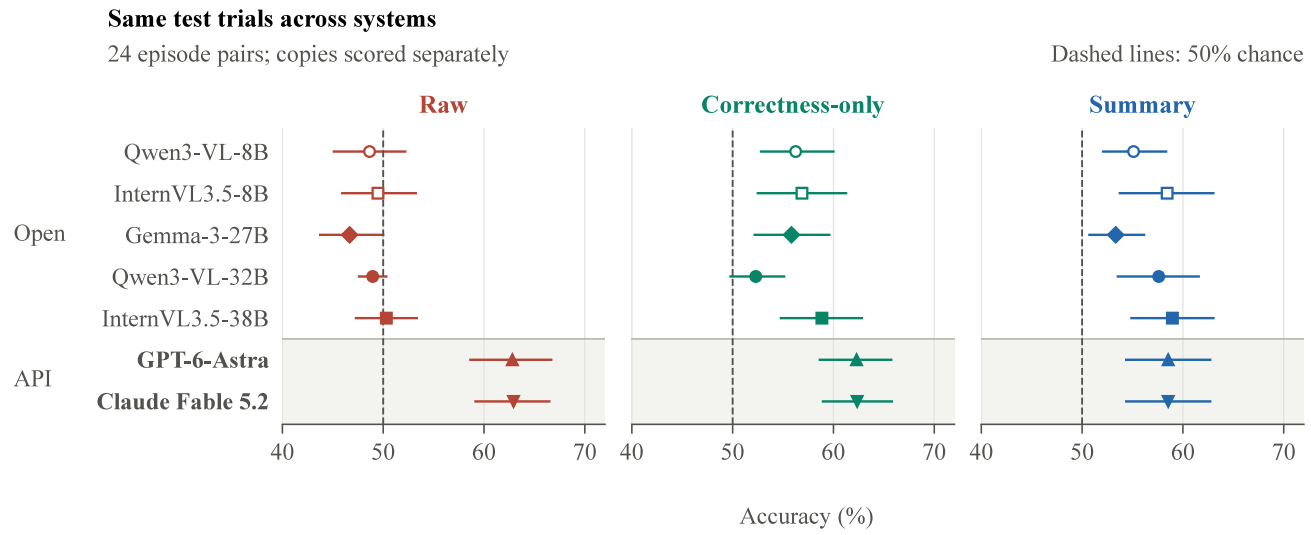


Figure 2: History-format effects differ across systems on the same held-out trials. Seven systems are compared on the non-pet cover stories of 24 held-out episode pairs, every fourth trial, with each counterbalanced copy scored separately. Open-model choices use the logits of the two answer words (candidate logits); API choices use generated answers (valid answers only). Each panel shows one history format. Bars show 95% bootstrap CIs from resampling episode pairs; dashed lines mark 50% chance. The comparison does not isolate reasoning from other system differences. Full-set results appear in Table 1.

more correct trials on 93.5–99.9% of non-tied trials, and four of five had accuracy equivalent to that of the all-trial count rule within ±2 percentage points (Gemma-3-27B: −1.4 percentage points, Holm-corrected equivalence-test $p = .053$; Table 5). A model that follows the count rule from SUMMARY therefore inherits that rule's response to change. The all-trial count rule can lag after reversals (Appendix B, Fig. 6): after a reversal, the mean following probability stayed below .5 early in the block with either SUMMARY or CORRECTNESS-ONLY, then rose above .5; with RAW it stayed near .50 (Fig. 4). Given the last-five and change-point SUMMARY variants, models agreed as closely with the corresponding count rules (choice agreement 94–100%; accuracy equivalent to that of the corresponding count rule in five of five models for the change-point variant and three of five for the last-five variant), and the mean following probability exceeded .5 two (change-point variant) or three (last-five variant) trials after a reversal; overall accuracy was 66.6–69.1%, 4–6 percentage points above SUMMARY and 4–8 percentage points above CORRECTNESS-ONLY (Appendix E). How fast an agent built this way adapts to an advisor whose accuracy changes is therefore set by the count rule outside the model.

Proprietary reasoning models learn from RAW. On a held-out subset (24 episode pairs, every fourth trial; Appendix J), Claude Fable 5.2 made correct choices on 62.9% [59.2, 66.5] of trials with RAW, 62.4% with CORRECTNESS-ONLY and 58.5% [54.4, 62.7] with SUMMARY (SUMMARY−RAW −4.4 percentage points [−8.6, 0.1]); GPT-6-Astra gave nearly identical values (62.8%, 62.3% and 58.5%; −4.3 [−8.8, 0.3]). On these same trials, with each counterbalanced copy scored separately, the five open models reached 46.7–50.3% with RAW, 52.3–58.9% with CORRECTNESS-ONLY and 53.3–59.0% with SUMMARY (Fig. 2); the Bayesian volatility learner reached 62.9% and the count rules 58.5% (all trials) and 67.9% (last five). CORRECTNESS-ONLY did not increase mean accuracy for the two proprietary models, and accuracy with SUMMARY was numerically lower than with RAW, at the level of the all-trial count rule. These models reason before answering (visible reasoning in 77% of Claude Fable 5.2's RAW answers) and also differ from the open models in scale, training and readout, so the comparison sets a boundary on the history-format effect rather than isolating an effect of reasoning. Their RAW choices were more consistent with advisor tracking than with the tested shortcuts (exploratory; Appendix J): they agreed with the Bayesian learner ($\kappa = .78$–$.87$) and the last-five count rule (.77–.79), not with repeating the previous outcome ($\kappa \approx 0$) or with the outcome-quota rule (.07–.10), which predicts the label with more occurrences left in the current block and would score 70.2% on these trials; where the outcome-quota rule and the Bayesian learner disagreed they followed the learner on 91–96% of trials, and after a reversal, 6–9% of their choices followed the currently better advisor in the first three trials of the reversed block, rising to 94–99% in trials 8–10. Given SUMMARY, both chose exactly as the all-trial count rule ($\kappa = 1.00$).

Image reports add context-dependent costs (exploratory). Models named a single report image almost perfectly, yet with described accuracy and image reports they chose the advisor described as better on only 59–84% of items, in four models least when the advisor's name was given only after the image (Appendix G). Control-item accuracy was mostly higher when the name was printed on the image or both reports were shown in one composite image (one labeled

Model	Raw	Raw+Correctness	Correctness-only	Summary	Summary−Raw [95% CI]	p_{Holm}
Full held-out set: 48 episode pairs; copies combined before scoring						
Qwen3-VL-8B	48.6	51.7	61.9	62.1	+13.4 [12.2, 14.6]	<.001
InternVL3.5-8B	47.9	50.8	63.6	63.5	+15.6 [14.5, 16.7]	<.001
Gemma-3-27B	47.0	49.8	60.2	61.7	+14.7 [13.2, 16.3]	<.001
Qwen3-VL-32B	46.8	52.2	62.8	63.1	+16.4 [15.0, 17.7]	<.001
InternVL3.5-38B	54.9	56.0	62.5	63.3	+8.4 [7.4, 9.4]	<.001
API subset: 24 episode pairs, every fourth trial; copies scored separately						
GPT-6-Astra	62.8	–	62.3	58.5	−4.3 [−8.8, 0.3]	–
Claude Fable 5.2	62.9	–	62.4	58.5	−4.4 [−8.6, 0.1]	–
Computational references: full set / API subset						
Bayesian volatility learner		66.8 / 62.9			–	–
Count rule (all trials)		63.2 / 58.5			–	–
Count rule (last 5)		69.2 / 67.9			–	–

Table 1: Correctness-only raises accuracy in the five open models. Accuracy is in percent; differences and their paired 95% bootstrap CIs are in percentage points. Full-set scores average the two cover stories after aligning and averaging the counterbalanced logits. API scores use generated answers on a smaller set of trials and are reported separately; Fig. 2 compares all systems on the same trials with per-copy scoring. Primary p-values are Holm-corrected across the five open models; secondary tests and controls appear in Appendix C; multimodal checks appear in Appendix G.

	Outcome label shown		Outcome label masked	
Model	Reports shown	Reports masked	Reports shown	Reports masked
Qwen3-VL-8B	51.7	52.0	55.7	60.4
InternVL3.5-8B	50.8	50.4	56.1	59.6
Gemma-3-27B	49.8	49.9	55.0	60.5
Qwen3-VL-32B	52.2	51.8	58.6	60.7
InternVL3.5-38B	56.0	55.7	58.6	63.3

Table 2: Masking past outcome labels improves accuracy; masking reports as well yields further gains. Accuracy (%) on 48 held-out episode pairs with text reports and counterbalanced logits combined before scoring. Columns specify whether past outcome labels and advisor reports are shown or replaced by a dash; correctness statements remain visible in all conditions. The first column is Raw+Correctness. Paired changes and their 95% bootstrap CIs appear in Fig. 11.

panel per advisor), although each panel was read correctly. On the held-out set, much of the cost came from choosing the first-listed image: with Correctness-only, four models chose the first-listed report on 79–92% of trials with images but 44–54% with text, and their accuracy fell to 11–24% when the correct report was listed second (Appendix I). With Correctness-only, image reports lowered held-out accuracy to 52.3–55.2%, 7–10 percentage points below text reports. Models could not read single cells of the correctness chart reliably (cell-reading accuracy 61–71%), so chart histories are not interpreted.

Humans and open models in the changing bandit. In the changing bandit of Psych-101, a normative bandit learner forgets the unchosen machine faster when the hazard rate is higher, and participants did (median forgetting rate .079 at $h = 0.1$, .255 at $h = 0.4$; Appendix H). Unlike the advice-taking task, this task varies the hazard rate across games. The five open models replayed participants' histories; participants had also played earlier games, so the comparison is descriptive. A simulation of the whole pipeline gave a false-positive rate of 6% but only about 50% power to detect a difference as large as the human effect (Appendix H), so a null result would not show similarity. Model estimates varied less with hazard rate. Their median forgetting rate was at most .051 at every hazard rate, and the median per-participant slope on hazard rate was -0.01 to 0.04, against 0.47 for the participants (Fig. 5). All five median paired differences were negative (-0.48 to -0.60), four at an uncorrected $p < .05$ ($p = .011$–$.070$), none after Holm correction across the five models. Stating the hazard rate, which matches what participants were told, did not produce a clear change (slopes -0.04 to 0.16). Forgetting rates near zero are estimated at the boundary of the parameter range, so we checked this in a separate regression: the models' choices did depend on the machines' last observed values (positive weight for 77–100% of participant histories), but whereas participants weighted a value difference less when the older value was stale (seen four or more trials earlier; stale/fresh weight ratio .58–.86), the models weighted stale differences as much or more (.95–2.07; Appendix H). This exploratory contrast does not establish a hazard-dependent adaptation deficit: the participants' stale-value discounting did not increase with hazard rate, and replay differs from acting in the task.

Discussion

Across five open VLMs answering directly, history format strongly affected how well models used advisor accuracy. The models used described accuracy and agreed with the all-trial count rule when given Summary, yet four of five chose at or below chance from Raw. Correctness-only recovered most of the Summary−Raw gap, Raw+Correctness helped little, and masking showed that the visible out-

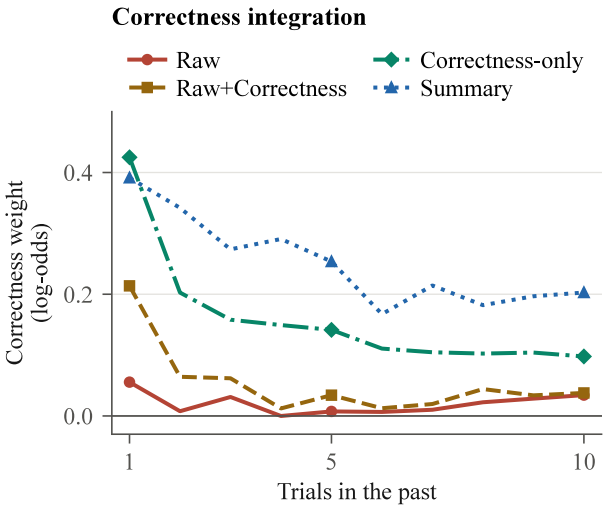


Figure 3: Correctness-only and Summary elicit decaying weights on past advisor correctness. Curves show integration kernels by history format, averaged over the five open models on held-out episode pairs with text reports. The first 10 of 15 fitted lags are shown; line styles, markers and colors identify formats. Per-model integration kernels and CIs appear in Fig. 8, and Raw strategy agreement in Fig. 9.

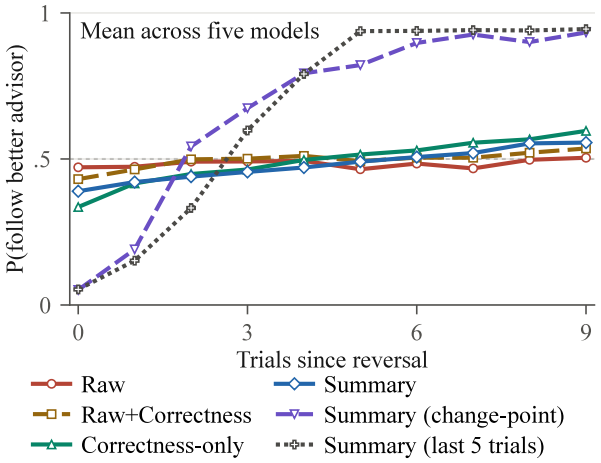


Figure 4: Last-five and change-point Summary variants recover higher following probabilities sooner after a reversal. Zero marks the first trial after a reversal. Curves are mean following probabilities over the five open models on held-out episode pairs, with line styles and colors identifying history formats. The dotted horizontal line marks .5; pointwise uncertainty is not shown. Per-model curves appear in Fig. 7.

come label elicited repetition of the previous outcome. This suggests a refinement of the description–history gap (Ge, Zhang, and Vorobeychik 2026): the reports and outcome labels in Raw promote a competing outcome-repetition response, even when trial-level correctness is explicitly stated in Raw+Correctness. Our manipulations change what the history states, not internal stages, so they do not localize a failed internal computation. The effect is also not universal: two proprietary reasoning models had observed Raw accuracy similar to that of the Bayesian volatility learner, with no increase in mean accuracy under Correctness-only. In the human study, who was correct is the observation from which a prediction error about the advisor is computed (Behrens et al. 2008); our interventions change how this observation is represented and make no claim about an internal error signal or its neural implementation.

Implications for NeuroAI. For building agents, an explicit record of which source was correct, and an integration rule outside the model (here, a count rule), help models that struggle with Raw: with the last-five Summary variant, mean following probability exceeded .5 three trials after a reversal; with the all-trial Summary it crossed .5 six trials after the reversal and reached only .56 at the end of the ten-trial window (Fig. 4). For models that already use Raw, Summary can instead remove information. For neuroscience, the format manipulation offers a way to test candidate models of advice taking against the behavioral signatures used for people; the open VLMs tested here do not yet show those signatures. These parallels are computational. In the changing bandit, participants' estimated forgetting increased with hazard rate, while model estimates varied less even when told the rate; the paired differences were not significant after correction. The two tasks probe complementary demands of adaptive inference, extracting evidence about advisor correctness and discounting stale information, and we do not assume that they share one mechanism.

Limitations. The task simplifies Behrens et al. (2008): two perfectly opposed advisors, no separate reward stream, and no human data on this variant. It is binary, with fixed outcome quotas, which allow shortcuts and make past outcome labels informative about future ones, information that Correctness-only removes; we therefore repeat the primary test on the robustness set (independent outcomes), where the Summary−Raw gap was about half as large (+3.6 to +8.3 percentage points) but significant in every model (Appendix I). The advice-taking task did not reliably detect volatility-dependent updating in the Bayesian volatility learner. We test this only in the changing bandit, where the human comparison is descriptive: participants were told the hazard rate and had played earlier games, and the test has about 50% power to detect an effect of the human size. Summary discards trial order, and the size of the Summary−Raw gap depends on the history's layout (Appendix I). The open models (up to 38B, four of five built on Qwen3) were prompted to answer directly; the proprietary models were tested on a subset and differ in several respects at once. In the interpretable advice-taking-task comparisons, images appear only as current reports. The analysis plan was fixed by a timestamped commit; deviations are listed in Appendix J.

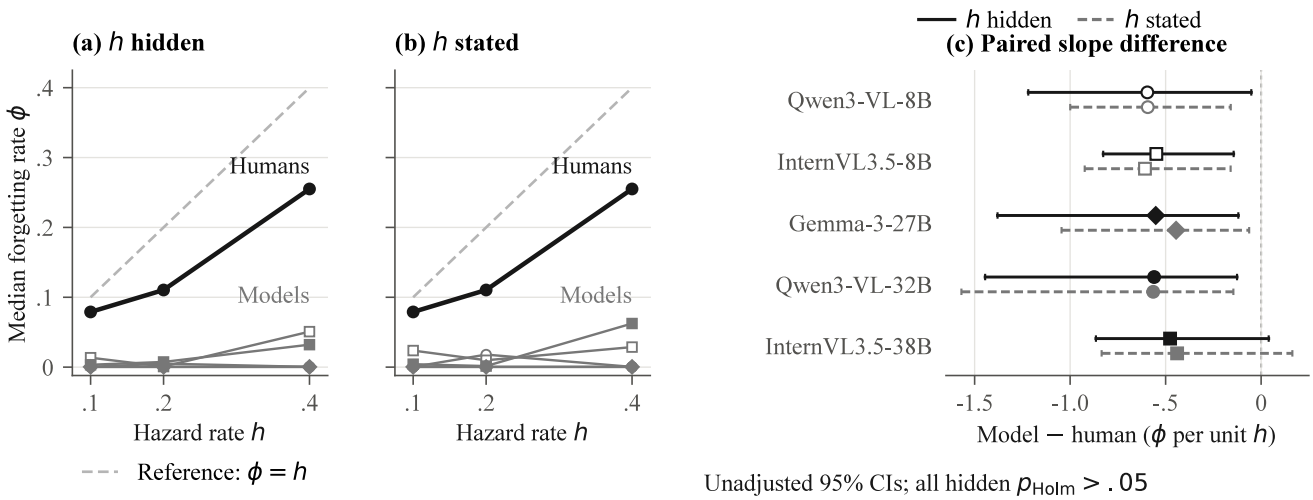


Figure 5: Comparisons between humans and the five open models remain descriptive. (a,b) Median fitted forgetting rate ϕ by hazard rate, with h hidden or stated for the model; humans were told h in both panels. Model estimates are medians over 20 fits to sampled binary-choice sequences. (c) Paired model-minus-human slope differences for the same 30 participant histories, with unadjusted 95% CIs from resampling participants. Solid and dashed intervals mark hidden and stated h; marker shape and fill identify the five models. None of the five hidden-h comparisons is significant after Holm correction. The simulation study shows limited power and an upward bias (Appendix H); these results do not establish equivalence.

Conclusion. Whether a VLM learns whom to trust from feedback depended on how the feedback was written down. For five open models answering directly, RAW invited repeating the previous outcome, and CORRECTNESS-ONLY recovered most of what SUMMARY provided; two proprietary reasoning models learned from RAW directly. For agents that must track changing source reliability, the history format and the integration rule are therefore part of the system to be evaluated, and the human-derived task and its computational references offer a way to test them against the behavior expected of people.

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Supplementary Material

The appendices provide exact task inputs, analysis definitions and checks of alternative explanations. Development-only checks are marked separately from held-out evidence; pending analyses are not presented as results.

A. Prompts and Cover Stories

Every prompt has four parts: instructions naming the two advisors and the cover story, the history in one of the formats of the main text, the current reports, and the question. Prompts are passed through each model's chat template as a single user message. For InternVL3.5 the template adds the model's default system message (in Chinese) and the suffix "Answer the question using a single word or phrase." All answer words are single tokens in every tokenizer used.

Task details. Each episode is shown with two cover stories that share the same outcome sequence: a pet cover story, in which advisors say whether a hidden animal is a cat or a dog, and one of six other cover stories (colors, weather forecasts, stock predictions, sports predictions, medical readings, news claims), assigned in groups of episodes so that cover story is crossed with phase order. Cover stories change only the wording, the answer words and the report images; all outcome sequences are synthetic. Current reports are text or an image: a programmatically drawn icon without text for the non-pet cover stories, and a photograph from the Oxford-IIIT Pet dataset (Parkhi et al. 2012) for the pet cover story (with text reports the pet cover story is text only, like the others). Both advisors' correctness is revealed on every trial (full feedback). In the counterbalanced copy, advisor names are swapped, the answer-word mapping and the advisor-introduction order are reversed, and the outcome sequence and images are unchanged; within each trial the two reports are listed in random order. The instructions say that an advisor's accuracy may change over time but give no rate. The held-out set has eight episodes per non-pet cover story.

Table 3 lists the cover stories. The examples below are rendered by the experiment code from the fixed episode files; line breaks are as in the prompt and *[image]* marks where an image is inserted.

Example 1: RAW, weather, held-out episode, trial 6.

> You are observing a sequence of days. On every day, two weather forecasters, Tim and Joe, give conflicting forecasts for the day's weather (rain or dry). After each day, the actual weather is observed. A forecaster's accuracy may change over time.
> Past days:
> Day 1: Joe forecasts rain; Tim forecasts dry. Actual weather: dry.
> Day 2: Tim forecasts rain; Joe forecasts dry. Actual weather: rain.
> Day 3: Tim forecasts rain; Joe forecasts dry. Actual weather: rain.
> Day 4: Joe forecasts rain; Tim forecasts dry. Actual weather: dry.
> Day 5: Tim forecasts rain; Joe forecasts dry. Actual weather: rain.
> Current day 6 (actual weather not yet observed):
> Joe forecasts: dry
> Tim forecasts: rain
> Predict the actual weather on the current day. Answer with exactly one lowercase word: rain or dry.

Cover story	Advisors	Answers	Report image
color	reporters	red/blue	color patch
weather	forecasters	rain/dry	cloud, sun
stock	analysts	up/down	trend arrow
sports	commentators	win/lose	happy/sad face
medical	doctors	positive/negative	plus/minus
news	news sources	true/false	tick, cross
pet	reporters	cat/dog	photograph

Table 3: Cover stories with the selected icon set. Trials are called trials (color, pet), days, weeks, games, cases or stories. Icons are drawn by code and contain no answer words. Appendix G shows the original and selected medical and sports icons and identifies results obtained with the original set.

The other formats change only the history lines. In RAW+CORRECTNESS, trial 1 reads "Day 1: Joe forecasts rain; Tim forecasts dry. Actual weather: dry. (Joe was wrong; Tim was correct.)". In CORRECTNESS-ONLY it reads "Day 1: Joe was wrong; Tim was correct.". In SUMMARY the five history lines and the "Past days:" line are replaced by "Summary of all 5 past days: Joe was correct 0 times; Tim was correct 5 times.". In the correctness chart they are replaced by "Past days 1-5 are summarized in this chart (days run left to right in four lines of 20; for each day, a green square means that forecaster was correct, a gray square means that forecaster was wrong, an empty square means not yet played):" followed by the chart (448 px; one row per advisor in each line). For the judgment question, the current reports and prediction question are replaced by "Before day 6: which forecaster is currently more likely to be correct? Answer with exactly one name: Joe or Tim.".

Example 2: described accuracy, weather, development episode. The described-accuracy control replaces the history with the following passage (advisor names vary by item); the remaining instruction and query structure is unchanged:

> Summary of past performance: Jim was correct on 80% of past days; Ken was correct on 20% of past days.

The described-counts control replaces the preceding passage with "Summary of the last 10 days: Jim was correct 8 times; Ken was correct 2 times.".

Example 3: pet cover story with image reports, held-out episode, trial 4.

> You are observing a sequence of trials. On every trial, two reporters, Jon and Jim, give conflicting reports about a hidden animal. After each trial, the true animal is revealed. A reporter's accuracy may change over time.

Past trials:
Trial 1: Jon reports cat; Jim reports dog. True animal: cat.
Trial 2: Jon reports cat; Jim reports dog. True animal: cat.
Trial 3: Jim reports dog; Jon reports cat. True animal: cat.
Current trial 4 (true animal not yet revealed):
Jim reports: *[image]*
Jon reports: *[image]*
Predict the true animal on the current trial. Answer with exactly one lowercase word: cat or dog.

With text reports the images are replaced by the answer word.

Binding variants. The binding controls change only how the current reports of Example 3 are attached to the advisors (described accuracy, image reports). *Name on image*: the lines are as in Example 3, but each image carries its advisor's name printed in a banner above the picture. *Composite image*: the two report lines are replaced by "Reports (one panel per reporter, name above each panel): *[image]*", one image with two panels, each headed by an advisor's name. *Name after image*: each report is given as "*[image]* (the image above was reported by Jim)", so the attribution follows the image. The composite-read check shows only the composite image and asks "The image has two panels, each with a name above it. What animal is in Jim's panel?".

B. Episode Generation and Counterbalancing

Schedules. Each episode has 80 trials in eight blocks of 10. Advisor A is the advisor with the higher scheduled accuracy on trial 1. Stable-first episodes have a stable phase (A correct with probability 0.8) followed by a volatile phase with blocks at 0.2, 0.8, 0.2, 0.8; volatile-first episodes have a volatile phase with blocks at 0.8, 0.2, 0.8, 0.2 followed by a stable phase at 0.8. Every episode therefore has four reversals (trials 41, 51, 61, 71 in stable-first and 11, 21, 31, 41 in volatile-first episodes). Advisor B is always correct when A is wrong. For comparison, the advice schedule of Behrens et al. (2008) (Supplementary Information) has 120 trials: 75% correct on trials 1–30, alternation between 80% correct and 80% incorrect every 10 trials on trials 31–70, and 85% incorrect on trials 71–120. Our volatile phase matches their alternation; our stable phase does not.

Relation to the original human study. Table 4 identifies the simplifications of Behrens et al. (2008). Their participants learned both reward-based option values and the accuracy of a confederate's advice. We retain changing advisor accuracy but omit the separate reward stream, reward magnitudes and in-person confederate; the schedules also differ. This is an adaptation of the social-learning component, not a replication of the human experiment. No human data exist for our variant, so the Bayesian volatility learner is a computational reference, not human performance; Psych-101 provides the separate human comparison (Appendix H).

Stratification. Within each block with scheduled accuracy p, A is correct on exactly $10p$ trials (8 or 2) in random order, and each outcome label is the truth on exactly five trials. This keeps realized accuracy equal to the schedule in every episode. It also fixes outcome quotas, which allow a shortcut that ignores the advisors: predicting the label with more occurrences remaining in the current block's quota (which presupposes knowing the quota) reaches 66.1% accuracy on the development episodes. We therefore also use a robustness set (below).

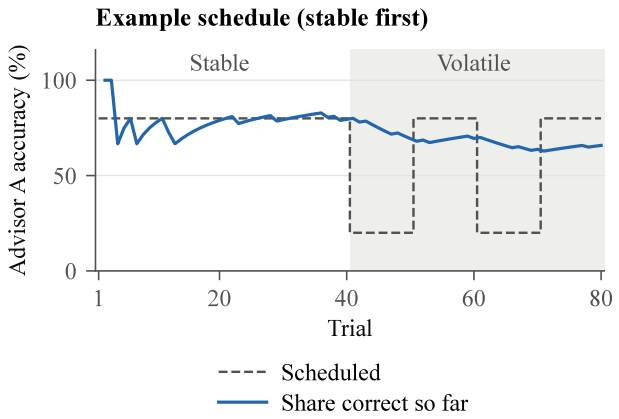


Figure 6: Why the all-trial count rule can lag after reversals. Scheduled accuracy of advisor A and its cumulative observed accuracy before each trial in one stable-first development episode. Shading marks the volatile phase; the cumulative count retains evidence from earlier blocks.

Names, order and counterbalancing. Advisor names are drawn from 16 common English first names that are single tokens in all tokenizers. The order in which the two reports are listed is random on each trial. Phase order alternates across episodes, the order in which the instructions introduce the two names alternates every two episodes, and non-pet cover stories rotate in groups of four episodes, so cover story, phase order and introduction order are fully crossed. Each episode has a counterbalanced copy with the advisor names swapped (A keeps its role), the answer-word mapping reversed and the introduction order reversed; the outcome sequence and images are unchanged. Analyses average the label-aligned logits within each episode pair.

Splits and seeds. The main set has 16 development episodes (color, weather, stock and sports cover stories) and 48 held-out episodes (eight per non-pet cover story), 128 with counterbalanced copies, generated with seed 20261111. Pet photos come from the Oxford-IIIT Pet dataset (CC BY-SA 4.0): 128 photos from the official trainval split for development episodes and 256 from the test split for held-out episodes, with no overlap. The robustness set has 24 episodes (48 with copies; seed 20261112) in which correctness is drawn independently on each trial, outcome labels are drawn independently, and volatile-phase dwell times (trials between reversals) are drawn uniformly from 6–16 trials. All data are regenerated from seeds and checked against stored SHA-256 hashes.

	Behrens et al. (2008)	This study
Agents	24 human participants (fMRI)	Five VLMs without fine-tuning
Choice	Blue or green option, each with a random reward magnitude	Which of two binary reports is correct (whose advice to follow)
Sources of information	Reward history of the options and one confederate's advice	Two advisors whose reports always conflict
Feedback	Correct option revealed after each trial	Outcome revealed after each trial
Advice schedule	120 trials; 75% correct, then 80%/20% alternating every 10 trials, then 15%	80 trials; 80% (stable, 40 trials) and 80%/20% alternating every 10 trials (volatile, 40 trials), both orders
Behrens-inspired Bayesian volatility learner on our episodes (trial-by-trial advisor correctness, no VLM)		
Accuracy	67.2% (development), 66.8% [66.1, 67.4] (held-out), 66.0% (robustness); all-trial count rule: 63.0 / 63.2 / 59.4%	

Table 4: The original human study and ours. The original advice schedule shows one counterbalanced orientation from the Supplementary Information of Behrens et al. (2008); its reward schedule is omitted. fMRI: functional magnetic resonance imaging. Accuracy of the Bayesian volatility learner: percentage of trials ($t \geq 2$) on which it favors the advisor who is correct, with a 95% bootstrap CI from 2,000 episode-pair resamples. Both copies have identical correctness sequences and learner predictions, so each pair is represented by its original episode.

C. Statistical Analysis, Control Conditions and Decomposition

Statistical analysis. The unit of analysis is the episode pair; all measures are averaged within an episode and its counterbalanced copy first, after averaging the label-aligned logits, which cancels additive name and label biases. Besides accuracy we report the following probability by phase, and following-probability curves aligned to reversals. The primary test, fixed before held-out evaluation (timestamped commit; not a formal preregistration), is the SUMMARY−RAW accuracy difference for text reports, per model, with a two-sided Wilcoxon signed-rank test and Holm correction across models; the decision criterion is a gap significantly above zero in both language-model families. RAW accuracy is called near chance only within 3 percentage points of 50%. Secondary tests are two one-sided tests (TOST; Lakens 2017) of whether a model given SUMMARY matches the count rule alone, with equivalence bounds of ± 2 percentage points; the position of CORRECTNESS-ONLY between RAW and SUMMARY (acc: accuracy), $\frac{\text{acc}_{\text{Correctness-only}} - \text{acc}_{\text{Raw}}}{\text{acc}_{\text{Summary}} - \text{acc}_{\text{Raw}}}$, interpreted only when the SUMMARY−RAW gap exceeds 5 percentage points and its CI excludes zero; and the primary test on the robustness set, as a separate family. Unless specified otherwise, accuracy and recovery-ratio CIs use 10,000 bootstrap resamples of episode pairs. Integration-kernel CIs use 1,000 such resamples. Intervals are pointwise 95% unless specified otherwise. We report per-story estimates and a sign test across the six non-pet cover stories (smallest attainable two-sided $p = .031$). Development episodes are used for exploratory checks and setting choices such as the window length K; held-out episodes are run once. The integration-kernel regression uses lags $k = 1, \ldots, 15$ plus regressors for repeating the previous outcome, a label bias and listing order; strategies for κ are repeating the previous outcome, win-stay/lose-shift (WSLS) on advisors (following the advisor correct on the previous trial, as in Behrens et al. 2008), the all-trial and last-five count rules, the BOCPD posterior-predictive rule and an outcome-label bias. These analyses describe behavior consistent with a strategy and do not identify a mechanism.

How much of the gap does CORRECTNESS-ONLY recover? On held-out episodes, CORRECTNESS-ONLY recovered 89–101% of the SUMMARY−RAW accuracy gap (Table 7). The ratio's uncertainty matters because a near-complete recovery is not an equivalence claim. Table 1 reports the accuracy means and paired SUMMARY−RAW differences; Table 7 supplies the individual accuracy CIs and the recovery ratio with its CI.

Control conditions. Table 6 gives each control condition with text and image reports, and Table 7 the fraction of the SUMMARY−RAW gap recovered by CORRECTNESS-ONLY. Image reports lowered control-item accuracy in almost every control condition, including described accuracy (59–84% instead of 92–100%); the one exception was Qwen3-VL-8B with a long easy history in RAW (53.3% vs. 53.1%). The image-report results use the selected icons (Appendix G). All selected icons were named correctly, and photo naming was 98.6–100%, so single-image recognition alone does not explain these drops. Control conditions were run on development episodes for all five models.

D. Rule Simulations and Bayesian Volatility Learner

We ran two simulations without any VLM to check the design. Both score a rule by the percentage of trials ($t \geq 2$) on which the advisor it chooses is correct; ties receive half credit.

Design simulation. The first simulation draws outcomes independently from the schedule (300 episodes per setting) and compares the BOCPD posterior-predictive rule (assumed hazard $h_{\text{CP}} \in \{1/8, 1/15, 1/30\}$), the last-$K$ count rule ($K \in \{3, 5, 7, 10, 15, 20, 30\}$), the all-trial count rule, and the reversal-aligned count rule. Across 20 settings (advisor accuracy 65–85%; phase lengths of 40/40, 60/60 and

Model	Count range	Model − count rule [90% CI]	$p_{\mathrm{TOST,Holm}}$	Choice agreement (%)
Qwen3-VL-8B	All trials	−1.1 [−1.7, −0.5]	.020	94.1
	Change-point	−1.2 [−1.7, −0.7]	.020	97.0
	Last 5	−2.4 [−3.4, −1.5]	.769$^{\times}$	94.3
InternVL3.5-8B	All trials	+0.3 [0.0, 0.6]	<.001	97.8
	Change-point	−0.5 [−0.7, −0.3]	<.001	98.5
	Last 5	−1.1 [−1.6, −0.6]	<.001	97.7
Gemma-3-27B	All trials	−1.4 [−1.9, −1.0]	.053$^{\times}$	93.5
	Change-point	−1.1 [−1.5, −0.6]	.002	96.5
	Last 5	−1.5 [−2.1, −0.9]	.088$^{\times}$	95.8
Qwen3-VL-32B	All trials	0.0 [−0.3, 0.2]	<.001	99.4
	Change-point	−0.3 [−0.5, −0.1]	<.001	99.3
	Last 5	−0.1 [−0.3, 0.0]	<.001	99.7
InternVL3.5-38B	All trials	+0.1 [−0.1, 0.3]	<.001	99.9
	Change-point	−0.1 [−0.1, 0.0]	<.001	100
	Last 5	0.0 [−0.1, 0.0]	<.001	100

Table 5: Secondary equivalence tests and agreement with the corresponding count rule on held-out trials. Differences and CIs are in percentage points and agreement in percent. The displayed 90% CIs are unadjusted bootstrap intervals (10,000 episode-pair resamples). Equivalence decisions use paired t-based TOST with the fixed $\pm$2-percentage-point tolerance and Holm correction over SUMMARY (all trials) and its two variants per model; $^{\times}$marks failure to establish equivalence. Last 5 was selected on development episodes for every model. Choice agreement is the percentage of non-tied trials with the same choice as the corresponding count rule; similar accuracy alone would not establish this. Primary SUMMARY−RAW tests appear in Table 1; all primary Holm-corrected $p < 10^{-8}$.

Condition	Reports	Qwen3-8B	IVL-8B	Qwen3-32B	Gemma-3-27B	IVL-38B
Described accuracy	Text	92.0	100	100	98.2	100
	Image	84.4	79.9	68.3	65.0	59.4
Described counts	Text	99.8	100	100	97.8	100
	Image	86.6	78.6	64.5	65.2	52.7
Short RAW	Text	70.3	63.2	60.7	60.0	78.6
	Image	64.1	54.2	51.8	59.2	54.0
Short CORRECTNESS-ONLY	Text	100	99.6	100	100	100
	Image	94.9	74.3	70.1	67.4	60.3
Long RAW	Text	53.1	56.7	55.6	54.5	69.6
	Image	53.3	52.0	42.4	51.6	46.2
Long CORRECTNESS-ONLY	Text	67.6	67.4	75.9	85.9	90.0
	Image	58.5	44.9	49.3	59.8	53.6
Single-trial check	Text	100	100	100	100	100
Described accuracy, name on image	Image	83.3	83.3	76.1	71.9	62.9
Described accuracy, composite image	Image	75.0	85.3	76.6	81.0	96.9
Described accuracy, name after image	Image	61.6	69.9	55.1	53.6	66.5
Composite-panel reading	Image	100	100	100	100	100
Correctness-chart reading	Chart	62.0	68.8	67.2	60.9	70.8

Table 6: Control conditions on development episodes (accuracy on control items in percent, averaged over counterbalanced copies and pooled over cover stories). Lower part: binding variants of the described-accuracy condition with image reports (advisor name printed on the image; both reports in one composite image; name given after the image), reading one panel of the composite image, and reading one cell of the correctness chart. Qwen3: Qwen3-VL; IVL: InternVL3.5. Image reports use the selected icons. Results by cover story will be included in the released analysis output.

80/40 trials; dwell times of 10 or 15 trials), the best BOCPD posterior-predictive rule minus the best last-K count rule ranged from -0.4 to $+0.8$ percentage points. For our setting (advisor accuracy 80%; phases of 40/40 trials; dwell time 10 trials), accuracy was 68.7% for the best BOCPD posterior-predictive rule ($h_{\mathrm{CP}} = 1/8$), 68.3% for $h_{\mathrm{CP}} = 1/15$, 69.0% for the best last-K count rule ($K = 3$), 63.2% for the all-trial count rule and 73.9% for the reversal-aligned count rule. Conclusions were the same for advisor accuracies from 65% to 85%, and accuracy varied by about one percentage point for assumed hazards h_{CP} between 1/8 and 1/30.

Count rules and model inputs. Table 8 applies the count rules corresponding to SUMMARY and its variants, together with hidden-information diagnostics, to our episodes. The last-five count rule outperforms the change-point count rule

Model	Accuracy 95% CI (%)				Fraction recovered
	RAW	RAW+CORRECTNESS	CORRECTNESS-ONLY	SUMMARY	[95% CI]
Qwen3-VL-8B	[47.3, 50.0]	[50.3, 53.2]	[60.6, 62.9]	[61.2, 62.9]	0.98 [0.90, 1.07]
InternVL3.5-8B	[46.8, 48.9]	[49.4, 52.1]	[62.8, 64.2]	[63.0, 63.8]	1.01 [0.96, 1.05]
Gemma-3-27B	[45.6, 48.3]	[48.4, 51.3]	[59.0, 61.2]	[61.0, 62.5]	0.89 [0.83, 0.96]
Qwen3-VL-32B	[45.4, 48.2]	[50.9, 53.6]	[62.1, 63.5]	[62.7, 63.6]	0.98 [0.93, 1.03]
InternVL3.5-38B	[53.8, 56.0]	[54.9, 57.0]	[61.7, 63.2]	[62.9, 63.7]	0.90 [0.81, 1.01]

Table 7: Uncertainty in the full-set accuracies and CORRECTNESS-ONLY recovery. The first four columns give pointwise 95% episode-pair bootstrap CIs in percent for the accuracies in Table 1, with counterbalanced logits combined before scoring. The last column is the dimensionless fraction of the SUMMARY−RAW gap recovered by CORRECTNESS-ONLY: $\frac{\mathrm{acc}_{\text{CORRECTNESS-ONLY}} - \mathrm{acc}_{\text{RAW}}}{\mathrm{acc}_{\text{SUMMARY}} - \mathrm{acc}_{\text{RAW}}}$. Recovery CIs are reported only when the gap exceeds 5 percentage points and its CI excludes zero, as fixed in the analysis plan.

Rule (no VLM)	Held-out	Robustness
Count rule (all trials)	63.2	59.4
Count rule (last 5)	69.2	67.5
Count rule (last 10)	61.2	63.5
Count rule (last 20)	57.9	60.4
Count rule (change-point)	67.8	66.3
BOCPD posterior-predictive rule	67.7	67.5
Count rule (reversal-aligned)	74.4	73.5
Always follow A	64.8	54.7
Known current advisor accuracy	80.0	78.8
Outcome-quota rule	65.5	–

Table 8: Rules on the 48 held-out episode pairs and the 24 robustness-set episode pairs. The change-point count rule uses the inferred run's majority; the BOCPD posterior-predictive rule uses BOCPD's predictive probability. Rows below the divider use hidden generator information and are hidden-information diagnostics, not available model strategies. The robustness set has no outcome quota. Held-out 95% bootstrap CIs (2,000 episode-pair resamples): last 5 [68.5, 69.8], change-point count [66.6, 68.8], all-trial count [62.8, 63.6]. Accuracies are in percent. Both copies yield identical rule choices, so each pair is represented by its original episode.

on both sets; the all-trial count rule lags after reversals, and the ordering of the longer windows differs across sets. We therefore do not claim that the change-point count rule is optimal or that these online rules are generally equivalent. The reversal-aligned count rule uses reversals only up to $t-1$, learning of each switch one trial late; it is not an upper bound. Knowing the current advisor accuracy is likewise not a strict ceiling because the main schedule has quotas (Appendix B).

These rule accuracies were computed on held-out episodes before any model results on the main data were seen. They are properties of the episode generator and involve no model behavior.

Bayesian volatility learner implementation. Our Bayesian volatility learner follows the volatility-learning idea of Behrens et al. (2007, 2008), using discrete Gaussian transition weights in logit advisor accuracy; the original formulation uses a beta transition in advisor accuracy. The grid has 60 equally spaced advisor-accuracy values $p \in [.01, .99]$, 20 equally spaced log-volatility values $v \in [\log(.02), \log(3)]$ (the log step size of changes in advisor accuracy), and a static, unknown meta-volatility $\eta \in \{.05, .2, .5, 1\}$, with a uniform joint prior. Before each trial, probability mass first moves across v with weights proportional to $\exp[-(v' - v)^2/(2\eta^2)]$, then across p with weights proportional to $\exp[-(\mathrm{logit}\, p' - \mathrm{logit}\, p)^2/(2e^{2v'})]$; each transition is normalized over destination grid points. Let $o_t = 1$ if advisor A was correct on trial t and $o_t = 0$ otherwise. The predictive mean $q_t = E[p_t \mid o_{1:t-1}]$ determines the choice before o_t is observed; a Bernoulli likelihood then updates the joint posterior, integrating over v and η. The learner chooses A when $q_t > .5$ and B when $q_t < .5$; ties receive half credit. The implied learning rate is $(q_{t+1} - q_t)/(o_t - q_t)$, excluding the last trial and prediction errors below 10^{-9} in magnitude. This finite-grid reference is not fitted to VLM choices.

E. Change-Point Controller and SUMMARY Variants

Change-point detection. Let $o_t = 1$ if advisor A was correct on trial t. Before trial t, the change-point controller (which selects the trial range for the change-point SUMMARY variant and count rule) runs Bayesian online change-point detection (Adams and MacKay 2007) on $o_1, \ldots, o_{t-1}$ with a Bernoulli likelihood, a Beta(1,1) prior for each run and a constant assumed hazard $h_{\mathrm{CP}} = 1/15$. This rate lies between the reversal rates of the volatile (1/10) and stable phases and was not tuned (Appendix D). The controller takes the maximum a posteriori run length r (the inferred number of trials since the last change), sets it to at least 3, and summarizes trials $\max(1, t - r)$ to $t - 1$.

SUMMARY variants. All variants use the same sentence as SUMMARY, restricted to a range of trials ("Summary of the last n days (days s–$(t-1)$): Ken was correct x times; Sam was correct y times."). The ranges are: all past trials (SUMMARY itself); and, as variants, the last $K \in \{5, 10, 20\}$ trials; the run since the last change point inferred by BOCPD (change-point variant); and the trials since the last true reversal up to $t - 1$ (reversal-aligned SUMMARY variant). Two further conditions truncate RAW instead of summarizing it: to the change-point run, or to the last 10 trials. The window length used in the secondary tests is, for each model, the one with the high-

est accuracy on development episodes, fixed before held-out evaluation; Table 9 reports the additional variants, and Table 5 gives rule agreement and equivalence tests. The last five trials were selected for every model. On held-out episodes, model accuracy was close to that of the corresponding count rule for every SUMMARY variant: last five trials 66.7–69.1% (count rule alone 69.2%), change point 66.6–67.7% (67.8%), last 10 trials 60.5–61.6% (61.2%), last 20 trials 58.1–58.6% (57.9%). The model thus adds little to, and takes little from, the count rule corresponding to its input; the choice of trial range determines that rule's accuracy. The same holds for the reversal-aligned SUMMARY variant (72.7–74.4%; count rule alone 74.4%). Truncating RAW did not help: with only the last 10 RAW trials, accuracy was 39.1–46.7%, at or below the full RAW history (46.8–54.9%) in every model, and truncating to the change-point run gave 43.5–50.2% across all five models. Shortening RAW did not recover the benefit of CORRECTNESS-ONLY.

What the controller is. The change-point controller is a training-free trial-range selector outside the model; it sees only the sequence indicating whether advisor A was correct and does not read model states. It is motivated by change-point models of human learning (Nassar et al. 2010, 2012), but we use it to select the trial range for correctness counts, not as a model of the VLM or of the brain.

F. Behavioral Diagnostics and Identifiability

What strategy agreement can establish. Cohen's κ adjusts agreement for chance. A rule that nearly always chooses one advisor can have high uncorrected agreement and low κ; matching a strategy does not identify an internal algorithm. Figure 9 shows selected RAW strategy comparisons, and Fig. 8 retains each model's kernels and CIs rather than only the mean in Fig. 3. Table 10 checks these diagnostics on synthetic agents: even a constant-rate delta learner can agree most with the BOCPD posterior-predictive rule, while its kernel has the expected geometric decay.

Pending model comparisons. Episode-grouped cross-validation fits remain pending and support no model-selection claim in this draft. Both cover-story versions of an episode must be assigned to the same cross-validation fold; fits that hold them out separately are not used. The reported strategy and kernel analyses do not depend on these fits.

Constant-rate delta learner. Let $o_t = 1$ when advisor A is correct. The delta learner updates $q_{t+1} = q_t + \alpha(o_t - q_t)$ from $q_1 = 0.5$, with learning rate α. Its choice probability is $P_t = \sigma[\beta(q_t - 0.5) + b]$, where σ is the logistic function, β the inverse temperature and b a bias. This defines the synthetic delta learner and the constant-rate null below.

Effective learning rate. The update of the model's choice probability in logit space after each trial, regressed on the signed correctness of advisor A on that trial separately for the stable and volatile phases, measures how strongly the model responds to new correctness observations. This statistic is not zero for a learner with a single constant learning rate: preferences are more extreme in the stable phase, where the logit stretches updates. We therefore calibrate it against a null in which a constant-rate delta learner fitted to the same episode generates simulated choice probabilities with the same residual noise (200 simulations); the calibrated volatile-minus-stable difference and its z score against the null indicate whether a model updates more in the volatile phase than a constant-rate learner would.

Parameter recovery. Simulations with constant-rate, phase-specific and confirmation-dependent delta learners (100 each; logit-noise standard deviation 0.3) recovered learning rates well but did not reliably identify the generating model: the Bayesian information criterion identified phase-specific learners in only 40% of cases, and confirmation-dependent learners in 10%. Thus parameter recovery alone does not justify an algorithmic interpretation of the VLM's behavior.

Can these episodes reveal volatility-dependent learning? We checked this with our Bayesian volatility learner, which allows volatility-dependent updating. On our 80-trial episodes its implied learning rate rose only from .100 in the stable to .115 in the volatile phase (held-out episodes). Neither the calibrated volatile-minus-stable effective learning rate ($+0.002$ $[-0.010, 0.014]$ over 16 development episodes) nor phase-specific integration kernels (volatile/stable ratio of the summed integration-kernel weights on the last three correctness observations: 0.84–0.90, versus 0.96 for a constant-rate learner) detected this change, so we do not use the advice-taking task to test whether models adapt their learning rate to volatility. The changing bandit (Appendix H), in which the hazard rate differs across whole games, tests that question instead.

G. Icon Design, Multimodal Binding and Chart Legibility

The legibility and binding controls use development episodes only and are exploratory; the comparison of text and image reports on the main task uses held-out episodes. All use the selected icons described below. Every model named the selected icons correctly. Pooled image naming was 98.8% for InternVL3.5-8B (98.6% for pet photos) and 100% for the other models, so recognition alone does not explain the remaining contextual costs (Table 11). In the described-accuracy control, control-item accuracy in four models was lowest when the advisor's name was given only after the image (54–70%) and higher when the name was printed on the image (72–83%) or both reports were shown in one composite image (75–85%), although each panel of a composite image was read correctly on every item; InternVL3.5-38B did best with the composite image (97%) and worst with plain image reports (59%). With CORRECTNESS-ONLY, image reports lowered held-out accuracy from 60.2–63.6% to 52.3–55.2%. Reading a single cell of the correctness chart was below the prespecified 80% chart-reading criterion for every model (61–71%).

Icon design and examples. Icons replace the answer word with a picture in the image-report conditions. Figure 10 shows the original and selected icon pairs at 224 pixels;

	Summary variants					Truncated Raw	
Model	Last 5	Last 10	Last 20	Change-point	Reversal-aligned	Last 10	Change-point
Qwen3-VL-8B	66.7	60.5	58.2	66.6	72.7	45.8	47.7
InternVL3.5-8B	68.1	61.3	58.6	67.3	73.7	46.7	47.2
Gemma-3-27B	67.7	60.9	58.1	66.7	72.8	39.1	43.5
Qwen3-VL-32B	69.1	61.5	58.2	67.5	74.0	42.9	45.3
InternVL3.5-38B	69.1	61.6	58.1	67.7	74.4	45.0	50.2
Count rule	69.2	61.2	57.9	67.8	74.4	–	–

Table 9: Summary variants and truncated Raw on held-out episodes (accuracy in percent); Summary appears in Table 1. The last-five-trial window was selected on development episodes for every model. The reversal-aligned Summary variant is a hidden-information diagnostic that uses true reversal times hidden from the models. Truncated Raw inputs have no separate count rule; change-point truncation uses the same inferred run as the corresponding Summary variant.

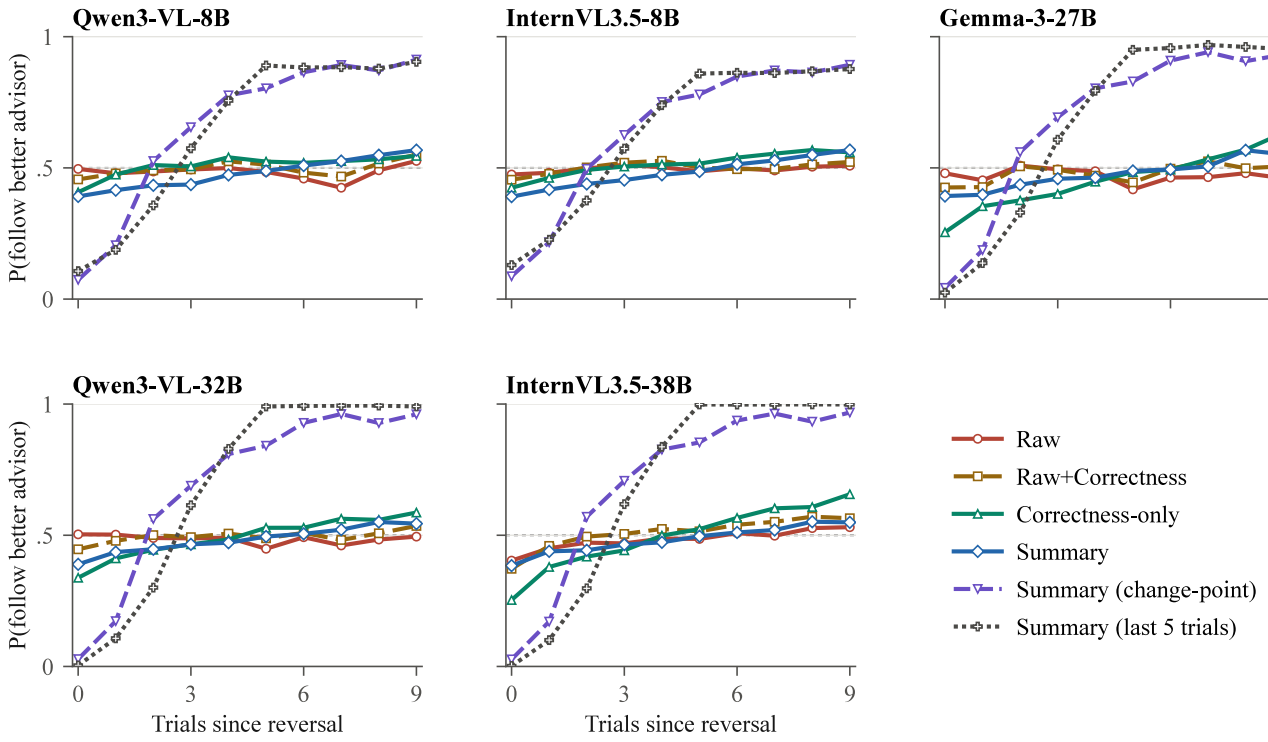


Figure 7: Following the currently better advisor after a reversal, separately for each open model and history format. Zero marks the first trial after a reversal. Each curve is a mean following probability on held-out episode pairs; the panels are descriptive and do not show confidence intervals. Fig. 4 averages the same curves over the five open models.

InternVL3.5 receives 448-pixel renderings from the same drawing functions. In the original medical pair, two red test-strip lines indicate positive and one indicates negative. The selected pair uses a dark plus or minus inside the same circular outline. In sports, the original gold trophy and red cross on a gray ball encode win and lose; the selected pair uses a yellow happy face and a gray-blue sad face. Labels and selection markers in the figure are reader annotations and are absent from the report images. These are synthetic symbols for the answer words, not clinical scans or photographs of game outcomes. All other cover-story icons are unchanged.

Development-only selection. We tested five medical pairs and four sports pairs with Qwen3-VL-8B and InternVL3.5-8B before held-out evaluation. The single-image probe asks for each of the two labels under both answer orders (four prompts per pair and model). We required all four responses to be correct for both models, then selected the eligible pair with the highest mean accuracy over four image-report controls and both models. The controls use described accuracy, a short easy history in Raw, and short and long easy histories in Correctness-only. Each downstream cell contains 64 items after averaging label-aligned logits across the counterbalanced episode copies. This selected plus/minus and

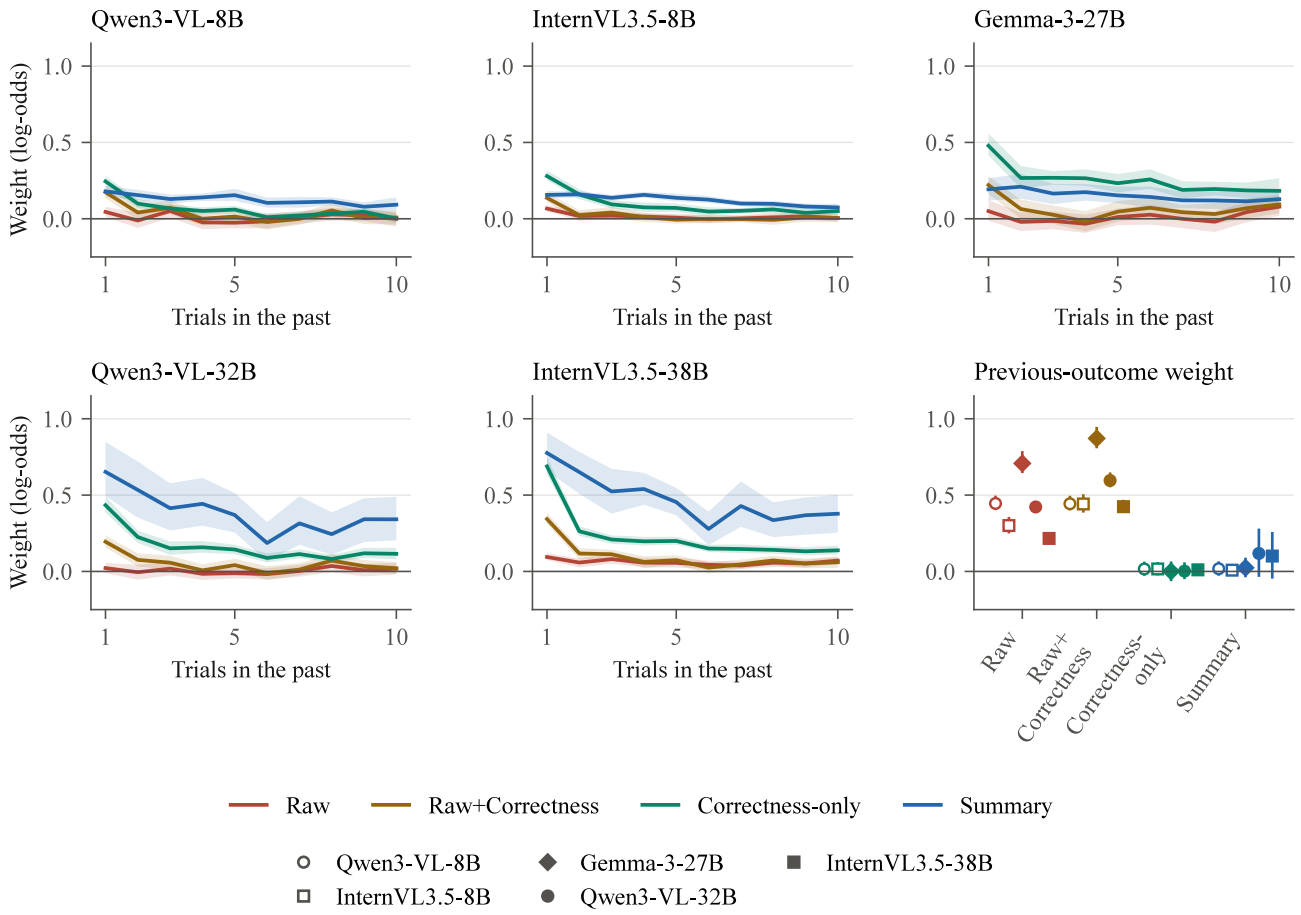


Figure 8: Integration kernels per model (shaded: 95% bootstrap CIs from 1,000 episode-pair resamples) and, bottom-right, the weight on repeating the previous outcome by history format (bars: 95% CIs; held-out).

Synthetic agent	Best strategy	κ	Repeat $\hat{w}$	Lag-1 $\hat{w}$
Repeat previous outcome ($w_{\text{gen}} = 3$)	Repeat previous outcome	1.00	3.00	0.00
Delta learner ($\alpha = 0.3$, $\beta = 8$)	BOCPD posterior-predictive rule	0.94	0.00	1.20
50% repeat previous outcome, 50% delta learner	Repeat previous outcome	0.49	1.21	0.56
Last-five count rule	Last-five count rule	1.00	0.02	1.08
All-trial count rule	All-trial count rule	1.00	0.05	0.27
Label bias ($w_{\text{gen}} = 3$)	Label bias	1.00	0.00	0.00

Table 10: Strategy and integration-kernel analyses applied to synthetic agents on the main episodes (development and held-out; no VLM). Best strategy: highest Cohen's κ. In agent labels w_{gen} is the generating weight; in column headings $\hat{w}$ is the fitted coefficient. Repeat $\hat{w}$ and Lag-1 $\hat{w}$: integration-kernel regression weights on repeating the previous outcome and on the advisor correct one trial back. For the delta learner the fitted integration kernel decays geometrically with ratio 0.700 (true $1-\alpha = 0.7$); the last-five count-rule agent has a flat integration kernel up to lag 5. Strategies are correlated on this schedule, so the delta learner agrees best with the BOCPD posterior-predictive rule; its fitted kernel nevertheless recovers the expected geometric decay. The same qualitative patterns held on the robustness set.

happy/sad faces. Enlarging the test strip left naming accuracy at 50% for both models, so size alone did not resolve that failure. Colored plus/minus signs scored higher downstream but failed the InternVL3.5 naming criterion.

Results and limits. The downstream mean increases from 52.1% to 66.8% for medical and from 58.8% to 67.8% for sports. Both selected pairs reach 4/4 in the naming probe for both models. However, accuracy in the described-accuracy condition, pooled over these two image-report cover stories

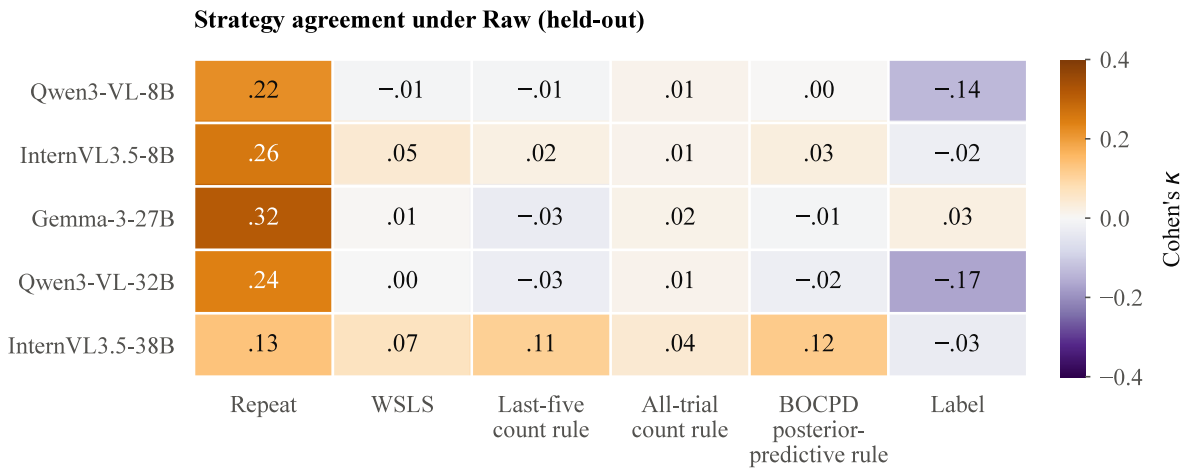


Figure 9: Under RAW, repeating the previous outcome agrees most with the open models. Cohen's κ on held-out episode pairs, with a color scale centered at zero. Repeat: repeat the previous outcome; WSLS (win-stay/lose-shift): follow the advisor correct on the previous trial; Last-five and All-trial count rules: choose by correctness counts over the specified range; BOCPD posterior-predictive rule: choose by its predictive probability; Label: outcome-label bias. Values are rounded to two decimal places; rounding zeros has no effect on their color.

	Image	Main-task reports	
Model	naming	Text	Image
Qwen3-VL-8B	100.0	61.9	55.2
InternVL3.5-8B	98.8	63.6	54.4
Gemma-3-27B	100.0	60.2	53.7
Qwen3-VL-32B	100.0	62.8	52.7
InternVL3.5-38B	100.0	62.5	52.3

Table 11: Image recognition versus use in context (in percent). Naming uses development items (photos and selected icons); main-task accuracy uses held-out episodes with CORRECTNESS-ONLY. Composite-panel and chart-reading controls appear once, in Table 6. Naming a report image does not establish successful use of that report in the task.

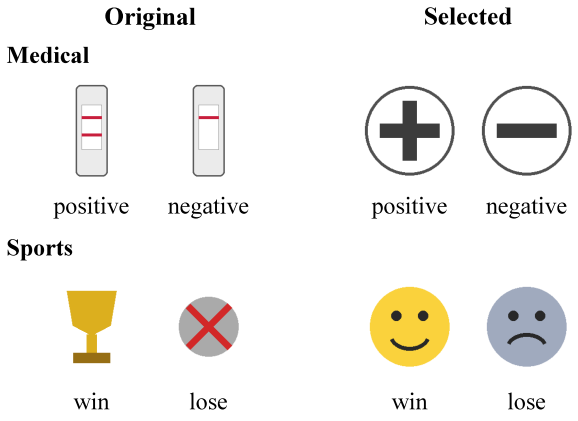


Figure 10: Original and selected report icons for medical (positive/negative) and sports (win/lose). Labels and headings are reader annotations and are absent from model inputs. Selection used development items only; the original recognition failures motivated the replacements.

is 77.3% for Qwen3-VL-8B and 72.7% for InternVL3.5-8B, below the 80% control-item accuracy criterion for the described-accuracy control. Single-image recognition therefore does not establish successful use of image reports in the advice-taking task. Selection and evaluation use the same development controls, so these gains are descriptive and do not estimate held-out improvement.

Versioning. Icon selection was a development-driven deviation from the fixed plan. Outputs with the original icons are kept separately and not reported. Text reports, pet photographs, the correctness chart and the primary text-report comparison are unchanged. The three models not used for selection also named the selected icons correctly. Tables 6 and 11 use complete reruns with the selected icons; the development gains above are not held-out evidence of improvement.

H. Human Comparison: Protocol and Validity Checks

Task and data. In the changing bandit of Xiong et al. (2023), participants repeatedly choose between two slot machines whose payoffs stay constant until a change point, which occurs with a hazard rate h; payoffs are noise-free, so a normative learner forgets the value of the unchosen machine at rate $\phi = h$. We use the 30 participants in Psych-101 (Binz et al. 2025) and their three 100-trial games ($h = 0.1, 0.2, 0.4$;

90 games, 9,000 trials). Participants were told h, and each 100-trial game directly preceded a 1,500-trial game with the same h, so by their second and third 100-trial games participants had played 1,600 and 3,200 trials; models see each game on its own.

Replay. Only the five open models are evaluated in this task; the proprietary models are not included. Models are conditioned on each participant's previously executed behavior (Binz et al. 2025): before trial t the model sees the original instructions and the participant's choices and payoffs on trials $1, \ldots, t-1$, and is asked which machine to press (one letter). The readout is the choice probability for each machine's answer letter, from the first-token logits normalized over the two letters, as in the main task. In the primary condition the hazard rate is removed from the instructions (hidden); in a secondary condition the original instructions state it (stated). Model and participant thus share the within-game history but not the hazard information (hidden) or the prior experience (both conditions).

Fits. We fit an adapted myopic model-free learner based on Xiong et al. (2023), separately for each hazard rate. It has free rates for updating the chosen machine's value and forgetting the unchosen one toward the midpoint (ϕ), plus choice sensitivity, perseveration and a bias. Humans give binary choices, so for comparability we draw 20 binary choice sequences from the model's choice probabilities in each game, fit each, and take the median. The primary statistic is each participant's slope of ϕ on h; the comparison is the paired difference between the model's and the human's slope (Wilcoxon test, participant-bootstrap 95% CI), with hidden- and stated-hazard conditions shown in Fig. 5. The human fits are our reanalysis; Xiong et al. (2023) report no fitted learning or forgetting rates by hazard rate. In our reanalysis, the median human ϕ rose from .079 ($h = 0.1$) to .110 and .255 ($h = 0.4$; per-participant slope 0.47 per unit h, positive in 73% of participants, Wilcoxon $p = .008$), adjusting for game position did not remove the increase (0.64 [0.23, 1.05]), and in the following 1,500-trial games the increase was weaker (median ϕ .076, .114, .131; slope test $p = .029$).

Calibration of the human–model comparison. We simulated the complete pipeline 200 times per condition. Simulated participants follow each real participant's fitted strategy at each hazard rate and play on the real value sequences; the simulated model follows the same strategy with a nominal forgetting-rate slope offset δ (rates clipped to $[.005, .95]$), and is replayed on the simulated participant's history; both are then fitted exactly as the real data (20 sampled choice sequences per model game, median fit, paired Wilcoxon test and bootstrap CI). With $\delta = 0$, the test rejected at $p < .05$ in 6.0% of replications (Monte Carlo standard error 1.7 percentage points) and the 95% CI covered the true difference in 93.5%, but the estimated difference was biased upward by about $+0.09$. Power was 0.49 for $\delta = -0.5$ and 0.59 for $\delta = +0.5$ (realized median slope differences -0.50 and $+0.39$ after clipping), with CI coverage 0.87 and 0.96. We therefore treat the comparison as supporting and descriptive (main text).

Image and chart payoff presentations. With the hazard rate hidden, the changing bandit is also run with the last five payoffs shown as pictures of the machine's display instead of text (224 px; 448 px for InternVL3.5), and with the whole history drawn as one scatter chart of payoff by trial for each machine (448 px). Each has a perception check: reading the number on one machine picture (four options; 450 items), and reading which machine was pressed, or which gave a higher payoff, at a given trial of the chart (540 items). These exploratory presentations are interpreted only when their perception checks succeed. Every model read the number on a machine picture (99.6–100%), but none read the chart reliably (which machine was pressed: 56–61%; which gave a higher payoff: 50–72%), so chart results are not interpreted. With picture payoffs, the sampled-choice fits also gave low forgetting rates (median at most .10; per-participant slopes -0.16 to 0.07). None of the five slope tests reached an uncorrected $p < .05$. Gemma-3-27B had the largest positive slope (.073, $p = .060$), with a paired model-minus-human difference of -0.23 $[-1.02, 0.21]$.

Regression check of forgetting (exploratory). For every trial on which both machines had been observed, we regressed the choice of the first machine (participants: binary; models: choice probability) on the difference between the two machines' last observed values (per 10 payoff points), separately when the older of the two values had been observed at most two trials earlier (fresh) or at least four trials earlier (stale), with the previous choice as a covariate (Table 12). All models were sensitive to value. We resampled the 30 participants as whole clusters (2,000 replicates), refitting the regression and using the same sampled participants for each human–model comparison. The stale-minus-fresh coefficient was negative for participants at $h = 0.1$ ($-.263$ $[-.501, -.114]$); its CIs included zero at the other hazard rates. Model-minus-human contrasts were positive at $h = 0.1$ and $h = 0.4$ in all five models (pointwise CIs above zero; Table 12). This supports a difference in stale-value use under replay, without establishing a hazard-dependent adaptation deficit.

I. Robustness, Prompt Wording and Masking

Robustness set and cover stories. On the robustness set, where correctness and outcome labels are drawn independently on each trial and reversals are irregular, RAW accuracy was 50.7–52.0% in four models (near chance) and 55.9% in InternVL3.5-38B, and SUMMARY reached 58.2–59.5%, close to the count rule on these episodes (59.4%). The SUMMARY−RAW gap was smaller than on the held-out set but positive and significant in every model ($+3.6$ to $+8.3$ percentage points, Holm-corrected $p < .001$; Table 13), so it does not depend on the outcome quotas. On the 48 held-out episode pairs the gap was positive in every non-pet cover story for every model (sign test $p = .031$, the smallest attainable) and in the pet cover story ($+8.0$ to $+17.9$ percentage points; Table 14); it was smallest for Gemma-3-27B with the medical and news cover stories ($+3.6$, $+4.7$).

Which part of RAW+CORRECTNESS prevents use of its correctness statements. RAW+CORRECTNESS recovered only

Agent (hazard hidden)	Stale − fresh value weight [95% CI] $h = 0.1$	$h = 0.2$	$h = 0.4$
Participants (hazard stated)	−.263 [−.501, −.114]	−.049 [−.170, .046]	−.040 [−.122, .034]
Qwen3-VL-8B	+.047 [−.037, .146]	−.009 [−.081, .080]	+.089 [.048, .133]
InternVL3.5-8B	+.100 [.052, .147]	+.057 [.016, .101]	+.087 [.054, .119]
Gemma-3-27B	+.049 [−.011, .110]	+.035 [−.016, .094]	+.053 [.007, .097]
Qwen3-VL-32B	+.040 [−.017, .102]	+.026 [−.047, .097]	+.099 [.052, .138]
InternVL3.5-38B	+.146 [.066, .229]	+.039 [−.020, .106]	+.105 [.070, .141]
	Model − human contrast in stale − fresh weight [95% CI]		
Qwen3-VL-8B	+.310 [.159, .552]	+.039 [−.063, .169]	+.129 [.049, .212]
InternVL3.5-8B	+.363 [.216, .598]	+.105 [.021, .217]	+.127 [.055, .208]
Gemma-3-27B	+.313 [.156, .541]	+.084 [−.007, .198]	+.093 [.014, .180]
Qwen3-VL-32B	+.303 [.139, .557]	+.075 [−.024, .189]	+.140 [.071, .214]
InternVL3.5-38B	+.410 [.236, .663]	+.087 [−.033, .225]	+.145 [.075, .226]

Table 12: Exploratory stale-value regression. Coefficients are log-odds per 10 payoff points of last-observed value difference. Negative stale-minus-fresh values indicate discounting of stale values. Pointwise 95% CIs use 2,000 participant-cluster bootstrap refits; the same participants are resampled for paired model–human contrasts. These are descriptive, unadjusted intervals, separate from the primary hazard-slope tests.

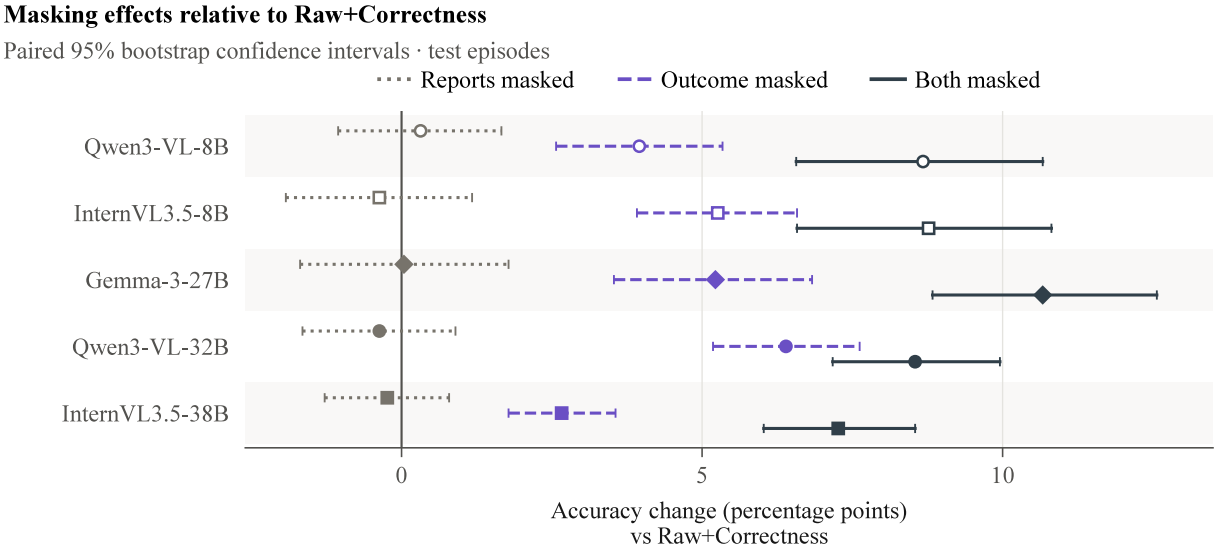


Figure 11: Paired masking effects corresponding to Table 2. Each contrast compares masking the reports, the outcome label, or both against Raw+Correctness; correctness statements remain visible. Markers show accuracy changes in percentage points, with pointwise 95% bootstrap CIs over the same 48 held-out episode pairs (10,000 resamples; text reports). The vertical line marks zero change. These are intervals for paired differences, not for the absolute accuracies in Table 2.

part of the Summary−Raw gap, and the weight on repeating the previous outcome stayed large. To find out why, we kept the layout and length of that history and replaced parts of each past trial with a dash ("Day 3: Ken forecasts —; Sam forecasts —. Actual weather: rain. (Ken was correct; Sam was wrong.)"): the outcome label, the two reports, or both. With both replaced the history carries the same information as Correctness-only at nearly the length of Raw+Correctness. On development episodes, replacing the outcome label raised accuracy by 7.4 and 9.6 percentage points (main effects; 95% CIs [5.2, 9.5] and [7.6, 11.5]) and lowered the weight on repeating the previous outcome from .48–.50 to .12 (Table 15), whereas replacing the reports had no effect on its own (+1.1 and −0.2 percentage points). With both replaced, accuracy was 1.5–2.1 percentage points below Correctness-only, leaving 14–20% of the original accuracy difference despite retaining much of the surrounding text; this comparison does not isolate additive contributions of length, layout and outcome labels. The outcome label thus elicits repeating the previous outcome even when who was correct is stated. Predictions for the held-out replication in five models were written before it was

Model	Raw	Summary	Gap [95% CI]	p
Qwen3-VL-8B	50.7	58.2	+7.5 [5.2, 10.0]	<.001
InternVL3.5-8B	52.0	59.4	+7.4 [4.2, 10.5]	<.001
Gemma-3-27B	50.7	59.0	+8.3 [5.5, 11.3]	<.001
Qwen3-VL-32B	51.6	59.3	+7.7 [5.1, 10.2]	<.001
InternVL3.5-38B	55.9	59.5	+3.6 [2.1, 5.1]	<.001

Table 13: Primary comparison on the robustness set (24 episode pairs with independent outcomes and irregular reversals). Accuracies are in percent and gaps in percentage points, with paired 95% bootstrap CIs over episode pairs. p: Wilcoxon signed-rank test, all Holm-corrected $p < .001$ within this family.

run. On held-out episodes, paired accuracy changes (Fig. 11) and weights on repeating the previous outcome (Table 15) supported all four predictions in all five models. Replacing the outcome label raised accuracy over Raw+Correctness ($+2.7$ to $+6.4$ percentage points; all $p < .001$) and cut the weight on repeating the previous outcome from .42–.87 to .08–.23, whereas replacing the reports showed no clear accuracy change (between -0.4 and $+0.3$ percentage points; all 95% CIs include zero) and left that weight at .26–.65. Averaged over the other factor, replacing the outcome label had a larger effect than replacing the reports in every model (difference of main effects $+2.9$ to $+6.8$ percentage points, all CIs above zero). With both replaced, accuracy was $+7.3$ to $+10.7$ percentage points above Raw+Correctness and within -4.0 to $+0.8$ percentage points of Correctness-only, despite retaining the longer layout. The held-out effect of replacing the outcome label alone is smaller than on development episodes and recovers 40–60% of the difference between Raw+Correctness and Correctness-only.

Prompt wording (exploratory, development episodes). We reran the four text history formats for Qwen3-VL-8B and InternVL3.5-8B with two alternative wordings; episodes, names, answer words, list order and readout were unchanged. Reworded sentences keep the original layout with different phrasing ("In trial 3, Sam said cat and Ken said dog; it turned out cat."; "Tally over the 3 previous trials: Ken was right 2 times and Sam was right 1 times."). A compact record uses one line per trial ("#3 | Sam=cat | Ken=dog | truth=cat"; "Correct counts so far (3 trials): Ken=2, Sam=1."). Table 16 gives the results. With reworded sentences, every result was preserved: Raw at 47–49%, Summary$-$Raw $+12$ and $+15$ percentage points, Correctness-only at or above Summary, Raw+Correctness recovering about 40% of the gap, and a positive weight on repeating the previous outcome. The compact record changed the picture in both directions. Raw accuracy rose to 55–56% and the weight on repeating the previous outcome fell, whereas Correctness-only and Summary fell to 56–60%, and InternVL3.5-8B showed only 72% choice agreement with the all-trial count rule when given Summary (98% with the original wording). The Summary$-$Raw gap shrank to $+3.4$ percentage points [0.8, 5.9] (Qwen3-VL-8B) and $+1.4$ [-1.9, 4.5] (InternVL3.5-8B). Raw accuracy stayed below the count rule (63%) in every wording, but the size of the Summary$-$Raw gap depends on how the history is laid out, and we interpret it only for the natural-language layout used in all four history formats.

Compact-record extensions to the three larger models gave Summary$-$Raw gaps of $+12.0$ percentage points [9.7, 14.6] (Qwen3-VL-32B), $+6.4$ [3.9, 8.7] (Gemma-3-27B) and $+7.1$ [5.1, 9.4] (InternVL3.5-38B), with paired 95% bootstrap CIs over the same 16 development episode pairs. Original-layout development runs are not all available on matched hardware, so these extensions do not isolate changes caused by layout. The compact record changes the task description, history notation and current-trial question together; this sensitivity check does not isolate which change is responsible.

Judgment question and report position (exploratory). On Raw trials the models were also asked which advisor was currently more likely to be correct (Appendix J). Table 17 compares the judgment averaged over counterbalanced copies with the choice on the same trials, and gives, for image and text reports with Correctness-only, the share of choices for the first-listed report and accuracy by the position of the correct report. Within each trial the two reports are listed in random order, identical in both counterbalanced copies. The models named the currently better advisor on 60–67% of Raw trials, below always naming the advisor that was better over most of the episode (75%), with little agreement with any rule ($\kappa \leq .17$), and their choices followed their own judgment on only 53–69% of trials ($\kappa \leq .12$): judgments and choices were weakly aligned.

J. Models, Readouts and Analysis Transparency

Table 18 lists the Hugging Face checkpoints and pinned revisions.

Open-model readout. The open models use bfloat16 with a float32 output projection. We extract first-token answer-word logits before processing and compute y_t as defined in Methods. Decoding is greedy; reasoning is disabled where the chat template permits. The judgment question with Raw and text reports asks "which forecaster is currently more likely to be correct?".

Readout quality. Table 21 reports full-vocabulary probability mass on the two answer tokens and greedy one-word answer compliance. For four models nearly all mass fell on the answer words in every history format. Gemma-3-27B's share of trials with candidate mass below 0.5 was at most .03 for Raw and Correctness-only, but .40 for Summary (.59 with photo reports), where it often began a sentence ("Based on the summary. . . "). On these trials its single-copy accuracy was 51%, versus 59% otherwise. Its Summary accuracy and failure of equivalence with the count rule may therefore understate how closely it would follow the all-trial count rule when given Summary and asked for a one-word answer; this readout limitation also constrains its primary Summary$-$Raw comparison.

Sensitivity to per-copy scoring (exploratory). We additionally scored each counterbalanced copy separately, then averaged over copies and cover stories within each episode pair. All 48 held-out episode pairs and trials $t \geq 2$ were

	Non-pet cover stories							
Model	Color	Weather	Stock	Sports	Medical	News	Sign test p	Pet
Qwen3-VL-8B	+10.9	+9.5	+15.7	+13.1	+9.7	+13.4	.031	+14.8
InternVL3.5-8B	+17.2	+15.7	+13.6	+11.7	+10.1	+11.6	.031	+17.9
Gemma-3-27B	+16.0	+14.6	+17.1	+15.7	+3.6	+4.7	.031	+17.5
Qwen3-VL-32B	+16.9	+18.7	+19.3	+15.8	+16.8	+10.1	.031	+16.4
InternVL3.5-38B	+5.5	+10.1	+12.7	+8.7	+8.7	+6.8	.031	+8.0

Table 14: SUMMARY−RAW gap by cover story in percentage points (held-out; eight episodes per non-pet cover story). Pet: the pet cover story, shown in every episode alongside its other cover story (text reports here; photographs only in image-report conditions); it is not part of the sign test.

Model	RAW+CORRECTNESS	Reports masked	Outcome label masked	Both masked	CORRECTNESS-ONLY
Development episodes (exploratory)					
Qwen3-VL-8B	.50	.48	.12	.04	.03
InternVL3.5-8B	.48	.27	.12	.02	.01
Held-out episodes					
Qwen3-VL-8B	.44	.40	.08	.01	.02
InternVL3.5-8B	.44	.26	.10	.03	.02
Gemma-3-27B	.87	.65	.18	.04	.00
Qwen3-VL-32B	.60	.60	.13	.02	.00
InternVL3.5-38B	.42	.37	.23	.02	.01

Table 15: Weight on repeating the previous outcome (log-odds) under history masking. Column names specify the information removed; correctness statements remain visible. Development rows are exploratory; held-out masking predictions were fixed before those runs. Absolute held-out accuracies appear in Table 2 and paired accuracy differences with CIs in Fig. 11; they are not repeated here.

Model	Wording	RAW	RAW+CORRECTNESS	CORRECTNESS-ONLY	SUMMARY	SUMMARY−RAW [95% CI]	Repeat
Qwen3-VL-8B	Original	48.5	53.4	64.0	61.7	+13.2 [10.8, 15.3]	.46
	Reworded	48.8	53.3	62.4	60.7	+11.9 [9.3, 14.4]	.50
	Compact	56.3	56.1	58.0	59.7	+3.4 [0.8, 5.9]	.32
InternVL3.5-8B	Original	47.1	53.0	64.0	62.8	+15.7 [14.0, 17.4]	.28
	Reworded	47.3	53.8	64.4	62.7	+15.3 [13.3, 17.5]	.26
	Compact	55.0	55.9	59.3	56.4	+1.4 [−1.9, 4.5]	.13

Table 16: Prompt-wording check (development episodes, text reports; accuracy in percent, gaps in percentage points, averaged over counterbalanced copies, $t \geq 2$). Repeat: regression weight on repeating the previous outcome under RAW.

	RAW, text reports				CORRECTNESS-ONLY			
	Judgment	Choice	Choice follows		Chose first listed		Image: acc. when correct is	
Model	names better	follows better	own judgment	κ	Text	Image	first	second
Qwen3-VL-8B	63.3	52.3	57.9	.09	45.5	52.3	57.1	52.5
InternVL3.5-8B	67.4	52.4	52.7	−.02	52.6	80.3	84.7	24.3
Gemma-3-27B	60.2	53.1	57.6	.07	43.8	79.2	82.8	24.3
Qwen3-VL-32B	60.7	53.2	53.7	.00	53.7	91.8	94.1	10.5
InternVL3.5-38B	67.2	63.3	69.1	.12	52.4	90.3	92.0	11.4

Table 17: Judgment and report position (held-out; averaged over counterbalanced copies). Names/follows better: the currently better advisor (higher scheduled accuracy on that trial); always naming the advisor that is better over most of the episode scores 75%. Values other than κ are in percent. κ: agreement between judgment and choice. With RAW and image reports, the larger models chose the first-listed image on 88–93% of trials (text 57–58%).

retained without candidate-mass filtering; ties received half credit. Every SUMMARY−RAW gap remained positive (Holm-corrected $p < .001$; Table 19), although its size changed. This supports the history-format effect under per-copy scor-

Model	Checkpoint	Revision
Qwen3-VL-8B	Qwen/Qwen3-VL-8B-Instruct	0c351dd01ed87e9c1b53cbc748cba10e6187ff3b
InternVL3.5-8B	OpenGVLab/InternVL3_5-8B-HF	741a7d03020411e666c6109218ab71e08151ef86
Gemma-3-27B	google/gemma-3-27b-it	005ad3404e59d6023443cb575daa05336842228a
Qwen3-VL-32B	Qwen/Qwen3-VL-32B-Instruct	0cfaf48183f594c314753d30a4c4974bc75f3ccb
InternVL3.5-38B	OpenGVLab/InternVL3_5-38B-HF	7c830fc25e874f9a861087e3c70587f2b9b555f3

Table 18: Checkpoints and pinned revisions (resolved 2026-11-11). B in model names denotes billions of parameters.

Model	RAW	CORRECTNESS-ONLY	SUMMARY	SUMMARY−RAW	CORRECTNESS-ONLY − RAW
Qwen3-VL-8B	49.3	58.6	59.4	10.2 [9.0, 11.2]	9.4 [8.2, 10.5]
InternVL3.5-8B	49.0	61.0	61.9	12.9 [11.9, 13.9]	12.0 [10.7, 13.2]
Gemma-3-27B	47.1	56.2	55.9	8.8 [7.3, 10.2]	9.1 [7.7, 10.5]
Qwen3-VL-32B	48.3	58.4	62.2	13.9 [13.1, 14.6]	10.1 [9.1, 11.1]
InternVL3.5-38B	53.9	61.7	63.1	9.2 [8.4, 10.1]	7.8 [6.9, 8.7]

Table 19: Per-copy scoring on the full held-out set. Accuracy is in percent and paired differences in percentage points, with 95% CIs from 10,000 bootstrap resamples of 48 episode pairs. Each copy is scored before averaging; the original counterbalanced-logit estimates appear in Table 1.

			Agreement with strategy (κ)					Quota vs. learner	Since reversal	
Model	History format	Acc.	Repeat	WSLS	All-trial count rule	Last-five count rule	Learner	follows learner	1–3	8–10
Claude Fable 5.2	RAW	62.9	.02	.43	.12	.79	.87	96	6	94
	SUMMARY	58.5	.02	.05	1.00	.11	.13	82	50	49
GPT-6-Astra	RAW	62.8	−.03	.41	.09	.77	.78	91	9	99
	SUMMARY	58.5	.02	.05	1.00	.11	.13	82	50	49
Open models (5)	RAW	46.7–50.3	.08–.31	.00–.04	.01–.05	−.01–.06	.01–.08	59–80	47–56	43–51
	SUMMARY	53.3–59.0	.01–.02	.01–.07	.04–.68	.09–.16	.08–.17	67–81	47–51	51–58
Rule alone		accuracy: repeat previous outcome 45.8; WSLS 62.3; all-trial count rule 58.5; last-five count rule 67.9; Bayesian volatility learner 62.9; outcome-quota rule (hidden) 70.2								

Table 20: Behavior on the API subset (24 episode pairs, non-pet cover stories, $t = 4, 8, \ldots, 80$; per-copy choices for all models). Accuracy and proportions are in percent; agreement is Cohen's κ. Strategy definitions follow Fig. 9; Learner denotes the Bayesian volatility learner. Quota vs. learner: share following the learner when it disagrees with the hidden outcome-quota rule. Since reversal: proportion of discrete choices following the currently better advisor in trials 1–3 or 8–10 of a reversed block (offsets 0–2 or 7–9). Only true reversals are included; initial blocks are excluded. These windows match Fig. 4, which plots following probabilities rather than proportions of discrete choices.

	Candidate mass			One-word
Model	Mean	Min	Share < 0.5	answers
Qwen3-VL-8B	1.000	.422	.000	1.000
InternVL3.5-8B	.997	.177	.000	1.000
Gemma-3-27B	.868	.000	.125	.875
Qwen3-VL-32B	.996	.807	.000	1.000
InternVL3.5-38B	1.000	.684	.000	1.000

Table 21: Readout quality pooled over history formats and report types on held-out episodes. Candidate mass sums the two answer-token probabilities; one-word answers is the proportion of greedy outputs equal to an answer word. Values are rounded. Gemma-3-27B's failures concentrate in SUMMARY, its variants and correctness-chart histories, as detailed in the text.

ing, still using candidate logits rather than unconstrained generation.

Proprietary models. GPT-6-Astra and Claude Fable 5.2 are queried through their APIs with the same prompts as the open models (text reports, non-pet cover stories only), on a subset of held-out episodes and query trials: episodes 0–23 of the held-out set with their counterbalanced copies (four per non-pet cover story, half starting with each phase), every fourth trial ($t = 4, 8, \ldots, 80$), and RAW, CORRECTNESS-ONLY and SUMMARY, 2,880 calls per model. Both were queried on 2026-11-13 with the default sampling settings (Claude Fable 5.2 does not accept a temperature), reasoning effort 'low' (the lowest that GPT-6-Astra accepts; Claude Fable 5.2 always reasons) and at most 4,000 output tokens; Claude Fable 5.2 was queried through the Message Batches API. The served model identifiers were gpt-6-astra and claude-fable-

5-2 on every call. The choice is the answer word: either the whole answer has at most three words and contains exactly one of the two answer words, or its last line is one of them; all other answers are invalid. GPT-6-Astra answered with one word on every call (median 5 output tokens, including reasoning); Claude Fable 5.2 gave visible reasoning before a final-line answer on 47% of calls (77% with RAW; median 47 output tokens), and 6 of its 2,880 answers (0.2%) were invalid.

Behavior of the proprietary models (exploratory). On the same trials with per-copy scoring, we tested advisor tracking against shortcuts using strategy agreement, trials on which the Bayesian volatility learner and the outcome-quota rule disagree, genuine reversal windows, and a logistic regression of choosing advisor A's report on A's correctness at lags 1–10 and on repeating the previous outcome (Table 20). The outcome-quota rule uses the generator's hidden 5/5 outcome quota and is a diagnostic only. With RAW both proprietary models gave large, decaying weights to past advisor correctness (lag 1: 2.1 and 3.3 log-odds; lag 5: 0.7 and 1.4) and no reliable weight on repeating the previous outcome; the open models' weights on past advisor correctness were near zero and their weight on repeating the previous outcome positive.

Image preprocessing. Icons and photos are rendered at 224 px for Qwen3-VL models and at 448 px (one tile, no tiling) for InternVL3.5. For Gemma-3-27B, 224-px renders are resized by the processor to the fixed 896-px input (256 image tokens; pan-and-scan off). The correctness chart is rendered at 448 px for all models.

Software and hardware. Open-model held-out, robustness, Psych-101, control and window-selection evaluations used NVIDIA H200 GPUs (141 GB), one GPU per run, with PyTorch 2.8.0 (CUDA 12.8) and Transformers 4.57.1. The exploratory icon selection, prompt-wording checks for the two 8B models, and development masking checks used NVIDIA RTX 3090 GPUs with the same model revisions, precision and readout. The larger-model compact-record extensions used H200s with the held-out software stack. Each run stores the rendered prompt hash, unnormalized candidate logits, the probability mass on the two answer tokens, and a fingerprint of the configuration.

Code and data availability. Upon acceptance we will release the generator, seeds, prompts, runners and analyses. Scripts download Oxford-IIIT Pet (CC BY-SA 4.0) and the Psych-101 slice (Apache-2.0); third-party data are not redistributed. The original Xiong et al. (2023) data have no declared license and are used for analysis only. Generated episodes are verified by SHA-256.

Analysis plan and deviations. A timestamped commit fixed the plan before held-out evaluation; it was not formally preregistered. Prior observations comprised the 16-episode pilot, partial development controls from one model, rule-only held-out baselines, and unanalyzed software-check outputs from a few held-out items. Before held-out model runs, the evaluation hardware changed and a 72B model and a 32B reasoning model were dropped for compute reasons. Development checks prompted icon replacements (Appendix G). Masking, prompt-wording checks and the Bayesian volatility learner were added after the commit. Masking predictions followed development results but preceded held-out masking: this later-specified replication is separate from the primary test. API behavioral comparisons, judgment–choice and image-position checks, stale-value regression and per-copy scoring sensitivity were specified after inspecting held-out outputs and are exploratory.

An exploratory development-only test of an open reasoning checkpoint was completed but excluded because truncation, parsing sensitivity and sparse sampling limit interpretation.

Pilot disclosure. The earlier RTX 3090 pilot used fixed advisor-introduction order, removed in the current episodes. Outcome repetition, image-position effects and poor chart reading motivated the controls and readability criterion. These pre-plan outputs are excluded from main-task estimates and supply no held-out evidence.